Check for updates

Received: 4 May 2022 | Revised: 6 June 2022 | Accepted: 7 June 2022

DOI: 10.1002/cey2.256

REVIEW

CARBON ENERGY WILEY

# Progress and challenges of prelithiation technology for lithium-ion battery

Zhenyu Huang[1,2] | Zhe Deng[1,3] | Yun Zhong[1] | Mingkang Xu[3] | Sida Li[1] | Xueting Liu[1,2] | Yu Zhou[1,3] | Kai Huang[1,3] | Yue Shen[1] | Yunhui Huang[1]

[1]State Key Laboratory of Material Processing and Die & Mould Technology, School of Materials Science and Engineering, Huazhong University of Science and Technology, Wuhan, Hubei, People's Republic of China

[2]Battery & Acoustics Research Center, Jiangsu Jitri-Hust Intelligent Equipment Technology Co., Ltd., Wuxi, Jiangsu, People's Republic of China

[3]Battery & Acoustics Research Center, Hust-Wuxi Research Institute, Wuxi, Jiangsu, People's Republic of China

**Correspondence**

Yunhui Huang, Zhe Deng, and Yue Shen, State Key Laboratory of Material Processing and Die & Mould Technology, School of Materials Science and Engineering, Huazhong University of Science and Technology, Wuhan, 430074 Hubei, People's Republic of China.
Email: huangyh@mail.hust.edu.cn, zhedengunique@gmail.com and shenyue1213@hust.edu.cn

**Funding information**

National Natural Science Foundation of China, Grant/Award Numbers: 22179045, 5202780089

**Abstract**

Prelithiation technology is widely considered a feasible route to raise the energy density and elongate the cycle life of lithium-ion batteries. The principle of prelithiation is to introduce extra active Li ions in the battery so that the lithium loss during the first charge and long-term cycling can be compensated. Such an effect does not need to change the major electrode material or battery structure and is compatible with the majority of current lithium-ion battery production lines. At this stage, various prelithiation methods have been reported, some of which are already in the pilot-scale production stage. But there is still no definitive development roadmap for prelithiation. In this review, we first introduce the influence of prelithiation on electrochemical performance from a theoretical point of view and then compare the pros and cons of different prelithiation methods in different battery manufacturing stages. Finally, we discuss the challenges and future development trends of prelithiation. We aim to build up a bridge between academic research and industrial application. Some engineering problems in the promotion of prelithiation technique are extensively discussed, including not only the implementation of prelithiation but also some collateral issues on battery designing and management.

**KEYWORDS**

cycle life, electrochemistry, initial coulombic efficiency, lithium-ion battery, prelithiation

## 1 | INTRODUCTION

In the past few centuries, the large-scale use of fossil fuels has brought about unacceptable environmental crises, such as global warming and extreme weather. There is an urgent need to develop renewable clean energies like wind and solar energy, as well as electric vehicles, to reduce the consumption of fossil fuels.[1,2] Nevertheless, the development of these new energies largely depends on rechargeable batteries.[3–5] The intermittent and random nature of wind and solar energy requires rechargeable batteries to balance power supply and consumption. Electric vehicles need high-performance batteries to increase their cruising range and service lifespan.[6–11] At present, lithium-ion batteries (LIBs) are the most prevalent rechargeable batteries.

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2022 The Authors. *Carbon Energy* published by Wenzhou University and John Wiley & Sons Australia, Ltd.

Promoting the energy density and cycle life of LIBs is of great importance to further develop new energy industries and protect the global environment.[12,13]

A typical LIB is composed of cathode,[14,15] anode,[16] current collector,[17] separator, and electrolyte,[18] whose charge/discharge behavior is based on the transfer of Li ions between the cathode and anode. The amount of transferable Li ions determines the capacity of the battery. In the conventional configuration, all of the transferable Li ions originate from the cathode material, while the anode material does not contain Li ions before the first charge process (so-called "formation"). However, not all of the active Li ions in the cathode material are transferable in the following cycles. In the formation process, approximately 6%–15% of Li ions extracted from the cathode material are irreversibly consumed to form a solid electrolyte interface (SEI) on the surface of the anode,[19–21] resulting in a significant decrease in usable capacity. In addition, some active Li ions may become inactive due to SEI growth and lithium-plating during cycling, which causes performance decay.[22] To address these issues, researchers have developed an available technique to introduce extra active Li ions in the battery to compensate for lithium loss,[23–25] which is called prelithiation (Figure 1A).

Prelithiation does not change the major electrode material or battery structure. Therefore, it is widely considered a highly possible industrial route to enhance the electrochemical performance of LIBs.[26] However, there is still no definitive development roadmap for prelithiation. In this review, we first introduce the influence of prelithiation on electrochemical performance from a theoretical point of view and then compare the possible routes to realize prelithiation in the LIB industry. Finally, we discuss the challenges and future development trends of prelithiation.

## 2 | EFFECT OF PRELITHIATION ON ELECTROCHEMICAL PERFORMANCE

The effect of prelithiation on the electrochemical performance of LIBs can be described by the relative position of the potential–capacity curves of the cathode and anode (Figure 1).[27–35] Without prelithiation, some active lithium ions are consumed in the formation process, and thus the capacity of the cathode material cannot be fully utilized during reversible cycling.[36,37] In the figure of the potential–capacity curves, the vertical overlapped part of the positive and negative curves represents the capacity of the full cell. The overlapped part is reduced as the curve of the cathode shifts to the left compared to the curve of the anode after formation, indicating that the low state of charge (SOC) part of the

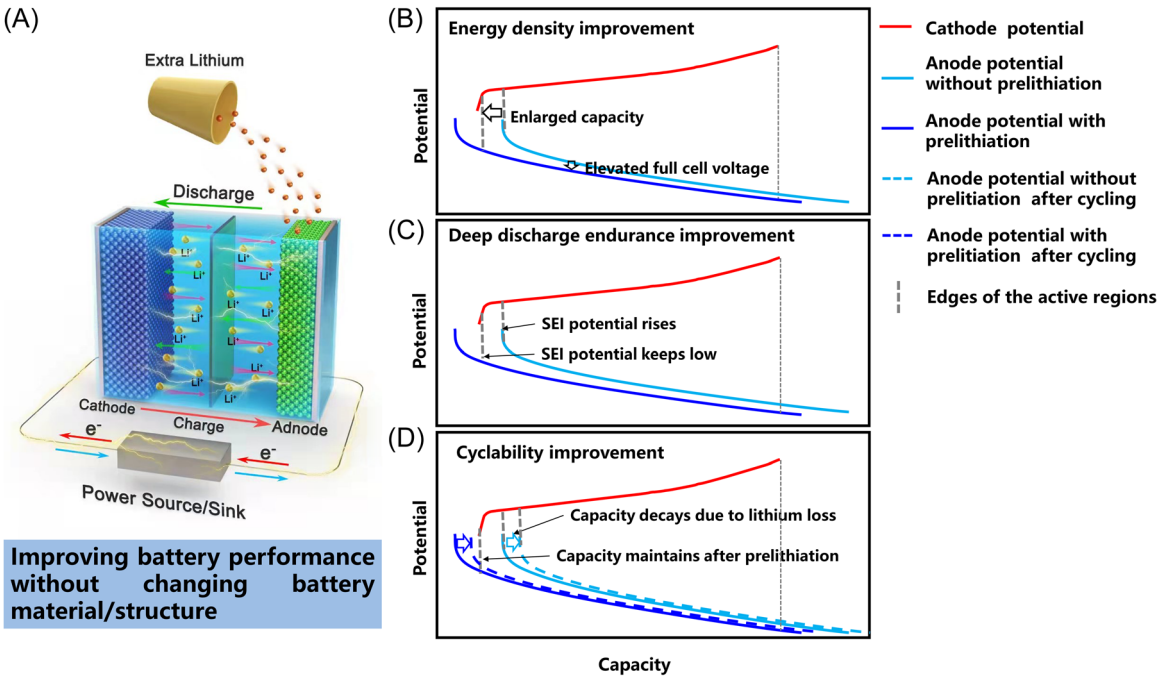



FIGURE 1 (A) Scheme of prelithiation technology. Benefits brought by prelithiation: (B) energy density improvement due to the increased capacity and elevated full-cell voltage; (C) deep discharge endurance improvements due to the low solid electrolyte interface (SEI) potential at the end of discharge; (D) cyclability improvement due to delaying the lithium loss effect.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

cathodic capacity is unexploited. In contrast, with prelithiation, as some extra lithium ions are introduced and inserted into the anode, the curve of the cathode shifts to the right, offsetting the shift to the left caused by lithium loss. In this case, the entire cathode curve overlaps with the anode curve, indicating that the total capacity of the cathode is utilized. Furthermore, by looking into the schemes of the potential-capacity curves, we can see that prelithiation may bring some extra benefits rather than just raising the specific capacity.

(1) Prelithiation elevates the full-cell voltage. As shown in Figure 1B, because of the relative shift in the anode potential–capacity curve, the average potential of the anode inside the overlapped section decreases, which enlarges the gap between the two curves. A larger gap between the overlapped curves indicates a higher full-cell voltage. This effect is especially important to making good use of anode materials with large potential changes at different SOCs, for example, Si,[32,38] $SiO_x$,[39] and metal oxides.[40]
(2) Prelithiation improves the SEI stability during deep discharge (Figure 1C). Without prelithiation, the full-cell voltage drop at the end of discharge is due to the increase in the anode potential. With prelithiation, the full-cell discharge ends due to the drop in the cathode potential. Such an electrode potential difference at the end of discharge may cause a significant difference in electrode stability. The SEI layer on the anode surface is formed by reduction reactions at low potential. The increase in anode potential may create a new reactive SEI-free surface and reductive electrolyte decomposition, which indicates the destruction and reformation of the SEI.[41,42] With prelithiation, the anode potential is relatively low, which helps to maintain the stability of the SEI in the low SOC region.
(3) Prelithiation increases the cycle life (Figure 1D). Active lithium loss is a major reason for the performance decay of LIBs. In the scheme of potential–capacity curves, lithium loss during cycling is reflected by the relative shift to the right of the anode curve. Without prelithiation, such a shift indicates that the additional cathodic capacity in the low SOC region is no longer active. In contrast, prelithiation causes the anodic curve to shift to the left of the cathode curve at the initial state. Although the curve of the anode still shifts to the right, the entire cathodic capacity remains functional as long as the extra lithium is not consumed. As a result, the negative effect of lithium loss is delayed by prelithiation.

In the LIB industry, it is generally believed that prelithiation can bring a 5%–10% energy density increase and a threefold to fivefold increase in cycle life. Thus, it is desirable to develop inexpensive and highly efficient prelithiation techniques to upgrade the current LIB manufacturing line.[43–52] In the following section, we will discuss the pros and cons of emerging prelithiation methods.

## 3 | INDUSTRIAL REALIZATION OF PRELITHIATION

In this review, the prelithiation methods are classified based on different battery manufacturing stages: the material preparation, electrode processing, and battery assembly stages (Figure 2). Prelithiation at each stage has its own pros and cons regarding the cost, lithiating efficiency, feasibility of mass production, and compatibility with existing production lines. Some of these methods have already been conducted at the pilot-plant scale, while some are freshly developed and have only been used in the laboratory. There is still no conclusion regarding which prelithiation method is the best. It is necessary to consider the processing details of the available methods and evaluate their compatibility with existing LIB manufacturing lines.

### 3.1 | Prelithiation during material preparation

Various Li-rich materials can be mixed with electrode materials in the slurry mixing process, including pure Li, $Li_2O$, $Li_3N$, Li alloys, and some overlithiated cathode materials. These Li-rich materials can be added to either the cathode side or the anode side. Any material with the capability to release large amounts of lithium ions in the electrolyte wetting or formation process can be viewed as potential prelithiation reagents. Unlike electrode materials, such a lithium-ion release process does not need to be reversible. Therefore, the range of available prelithiation reagents is broader than that of electrode materials. The most significant advantage of the direct material mixing method is that only the recipe of raw materials needs to be changed while the battery manufacturing process can be mostly maintained. Thus, the existing production equipment does not need to change very much. However, it should be noted that prelithiated electrode materials are usually very sensitive to humidity. Therefore, it would be necessary to upgrade the humidity controlling system of the LIB manufacturing plant. In the following section, we will introduce the recent developments of prelithiation methods at the material preparation stage, while also referencing different lithiated materials.

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

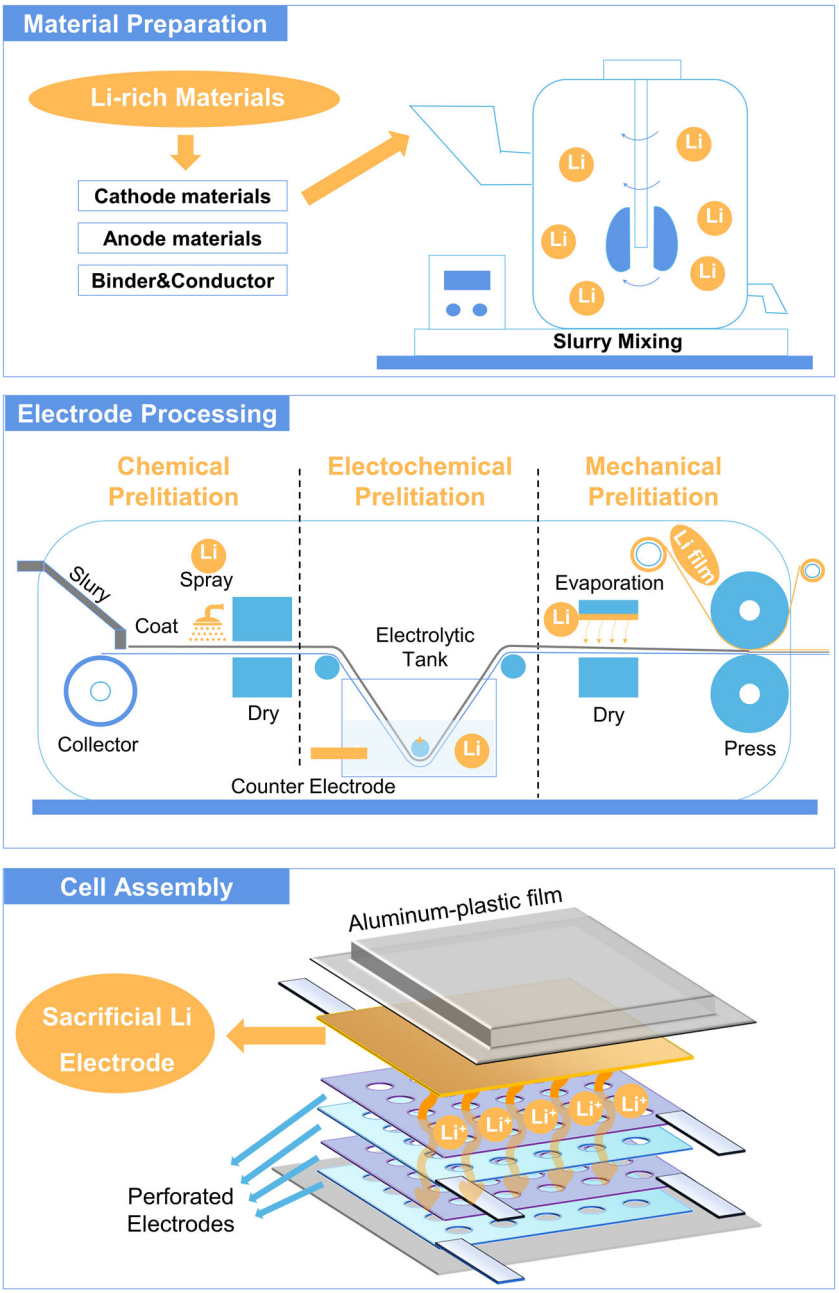



FIGURE 2 Classification of the prelithiation methods regarding the implanted battery manufacturing stages.

### 3.1.1 | Prelithiation of cathode materials

A common strategy for prelithiation is to introduce excess lithium during the synthesis of cathode materials. This excess lithium in the cathode can be released into the battery system during the first charge cycle.

One way to achieve cathode prelithiation is to directly insert excess lithium into the cathode material. As the lithium concentration of the cathode is higher than normal, this is also called overlithiation. The typical overlithiation method involves immersing cathode material powder into a reductive Li-containing reagent. In 1998, Peramunage et al.[53] reported that an overlithiated $Li_{1+x}Mn_2O_4$ cathode material could be synthesized by immersing spinel $LiMn_2O_4$ into a BuLi ($C_4H_9Li$) hexane solvent in an argon atmosphere. BuLi could react completely after stirring overnight, and the byproduct could be easily removed via subsequent filtration and rinsing with dry hexane. The charge–discharge test of the $Li_{1+x}Mn_2O_4$||Li ($x = 0$, 0.2, 0.3, 0.4, 0.6, and 1.0) cells revealed that the excess Li could be fully extracted from all these overlithiated compounds when charging, and the electrochemical cycling performance of $Li_{1+x}Mn_2O_4$|| graphite and $Li_{1+x}Mn_2O_4$||coke full cells confirmed that the excess lithium could compensate for the irreversible lithium loss in the first cycle. However, both BuLi and $Li_{1+x}Mn_2O_4$ are sensitive to humidity and oxygen, which makes it necessary to conduct the subsequent battery

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

manufacturing steps in an inert atmosphere; this is a great challenge for battery manufacturers. Liu et al.[54] reported an original idea to solve this problem. Instead of directly using overlithiated cathode materials, they coated conventional cathode materials with a stable layer that had a high lithium content. As shown in Figure 3A, $LiCoO_2$ was prelithiated via immersion into a lithium naphthalenide (Li-Naph, $Li^+C_{10}H_8^-$THF) tetrahydrofuran (THF) solution. The surface layer of the $LiCoO_2$ particles was in situ transformed into a $LiO_2$/Co nanocomposite via a chemical reaction ($LiCoO_2 + 3Li^+ + 3e- \rightarrow 2Li_2O + Co$, Figure 3B). This $Li_2O$/Co shell was stable in industrial electrode fabrication dry rooms and its thickness could be easily controlled by tuning the reaction time, making it possible to control the prelithiation degree. The initial charge capacity of the as-prepared $LiCoO_2$@$LiO_2$/Co (CS–$LiCoO_2$) cathode was 15 mAh $g^{-1}$ higher than that of pure $LiCoO_2$. When paired with the graphite–SiO anode, the CS-$LiCoO_2$||graphite–SiO cell showed a higher discharge capacity (141 mAh $g^{-1}$) than the conventional $LiCoO_2$||graphite-SiO cell (127 mAh $g^{-1}$). The cycling stability of the full cell was also improved. Notably, such a prelithiation method is simple and compatible with current battery fabrication processes, including slurry fabrication, electrode drying, and battery assembly.

However, limited by the maximum lithiation degree of the cathode materials, the overlithiation method can only achieve a low prelithiation degree. Moreover, overlithiation may have a negative effect on the reversibility of lithium-ion intercalation during operation. To overcome these drawbacks, a series of sacrificial Li-rich materials are physically mixed with the cathode materials; notably, these sacrificial materials do not change the structure of the cathode material. Nonmetallic Li-rich sacrificial compounds, such as $Li_3N$, $Li_2O$, $Li_2C_3O_5$, and $Li_2C_4O_6$, were first proposed by Michel Armand et al.[57] These compounds decompose

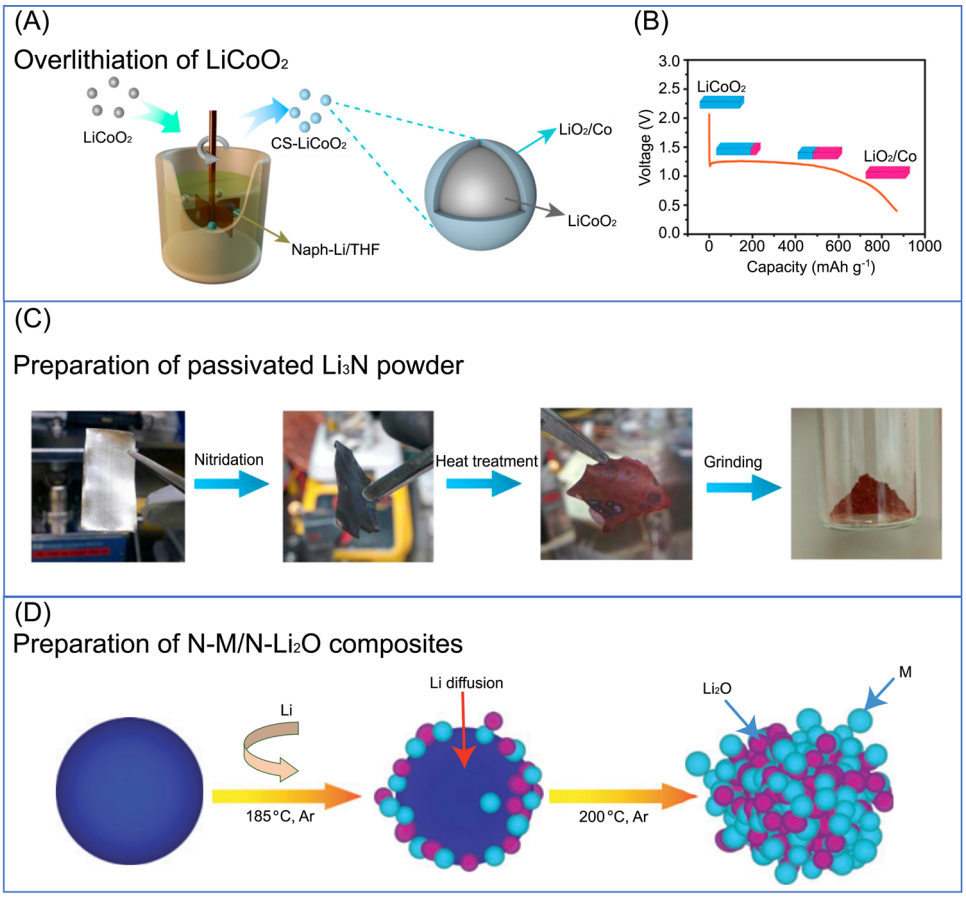



FIGURE 3 Prelithiation of cathode materials during their preparation: (A) schematic illustration showing the preparation of overlithiated $LiCoO_2$; (B) voltage-capacity plots of $LiCoO_2$ during its electrochemical lithiation process at 0.1 C (the inset illustrates the intermediate products during the conversion reaction of $LiCoO_2$). Reproduced with permission: Copyright 2020, American Chemical Society.[54] (C) Preparation process of the passivated $Li_3N$ powder. Reproduced with permission: Copyright 2017, Elservier.[55] (D) Schematic showing the fabrication process of N–M/N–$Li_2O$ composites. Metal oxides (MOs) are used as the starting materials and in situ converted into N–M/N–$Li_2O$ composites through a chemical reaction with molten Li. Reproduced with permission: Copyright 2016, Springer Nature.[56]

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and release active lithium ions during the first charge cycle within an acceptable potential range from 3.0 to 4.5 V, as shown in following formulas:

$$2LiN_3 \rightarrow 3N_2 + 2e^- + 2Li^+, \quad (1)$$

$$Li_2O \rightarrow 1/2O_2 + 2e^- + 2Li^+, \quad (2)$$

$$Li_2C_xO_y \rightarrow y/2CO_2 + (x - y/2)C + 2e^- + 2Li^+. \quad (3)$$

This type of sacrificial prelithiation material does not contain precious metal elements, while it often has a low preparation cost and a high lithium content. In particular, $Li_3N$ has the highest mass percent of lithium among all sacrificial salts. However, it is worth noting that pure $Li_3N$ demonstrates high chemical reactivity to moisture and is not compatible with the *N*-methyl-pyrrolidone (NMP) solvent, which is widely used in conventional slurry casting processes. To address this issue, Sun et al.[55] reported a simple way to prepare surface-passivated $Li_3N$ sacrificial salts by treating lithium metal with nitrogen gas and annealing in a glovebox at 200°C for 24 h with trace oxygen species (Figure 3C). A dense $Li_2O/Li_2CO_3$ passive layer was formed on the surface of the $Li_3N$ particles, which improved the stability of $Li_3N$ in ambient air.

Some Li-rich transition metal oxides ($Li_xMO_y$, M = Fe, Co, Ni, Mn, etc.) can also be suitable sacrificial additives on the cathode side. For example, $Li_2NiO_2$ was investigated and proposed by Park et al.[58] Their work confirmed that crystalline $Li_2NiO_2$ would transform into amorphous or trigonal $LiNiO_2$ phases and extracted $Li^+$ at 3.5 V (which is below the charging potential of common positive electrodes) during the first charge, acting as a satisfying sacrificial cathode additive. Moreover, $Li_6CoO_4$ with a higher lithium proportion than $LiNiO_2$ was investigated and reported by Noh et al. Their research showed that $Li_6CoO_4$ exhibited a high irreversible capacity ratio of 96% (318 mAh $g^{-1}$) during the first cycle. As it had a lower charging potential than $LiCoO_2$, it could effectively donate Li ions ahead of $LiCoO_2$ during the initial charge cycle. However, Co is relatively expensive. Therefore, $Li_5FeO_4$ with its lower cost compared to $Li_6CoO_4$ has been developed as a sacrificial additive. Su et al.[59] investigated the prelithiation effects of $Li_5FeO_4$ in a $LiCoO_3$||hard carbon cell. They found that after the addition of $Li_5FeO_4$, the capacity, output voltage, and energy density of the $LiCoO_3$||hard carbon full cell could simultaneously increase from 206 to 401 mAh $g^{-1}$, from 3.5 to 3.8 V, and from 300 to >400 Wh $Kg^{-1}$, respectively.

Another series of Li-rich sacrificial compounds, M–$Li_xY$ (M = Fe, Co, Ni, Mn, etc., Y = O, S, F), were recently reported by Sun et al.[56] As Figure 3D demonstrates, M–$Li_xY$ is fabricated by MY and molten metallic Li in an Ar atmosphere by a one-step conversion reaction. With such a cathode prelithiation additive, the specific and volumetric energy densities of a $LiFePO_4$/graphite full cell were increased by 8%–13% and 11%–14%, respectively. Furthermore, Liu et al.[54] chemically coated Co-$Li_2O$ nanoshells in situ on the surface of $LiCoO_2$ as a high-capacity built-in prelithiation reagent. Their experiments show that after 12-h ambient condition exposure, the cycling stability of the active $LiCoO_2$ in the CS-$LiCoO_2$ remains.

Compared to using overlithiated cathode materials, the addition of sacrificial Li-rich additives is more flexible. The degree of prelithiation can be easily tuned by adjusting the amount of the additive. In addition, the sacrificial additives decompose in the first charge cycle. Thus, they do not influence the electrochemical performance of the cathode material in the following cycles. However, the byproducts from the decomposition of the additive may result in some adverse effects on the batteries. If the byproduct is solid, such as the $Fe_2O$ produced from the delithiation of $Li_5FeO_4$,[60] it will stay in the cathode as an inert residue, which reduces the energy density. If the byproduct is a gas, such as $N_2$ released from $Li_3N$, the amount of gas released during formation would be much higher than that in conventional LIBs. Therefore, it would be necessary to keep the current at a very low level during formation and to introduce some advanced degassing techniques for preventing gas accumulation. Such a gassing problem is more serious when making large cells. To solve this problem, Lai et al.[61] reported lithiated 3, 4-dihydroxybenzoic acid ($Li_2DHBA$) as a sacrificial Li-rich compound. It is an organic salt synthesized through the reaction between LiH and 3,4-dihydroxybenzoic acid. Compared to other sacrificial compounds, $Li_2DHBA$ has an environmentally friendly manufacturing process. More importantly, the byproduct of such a prelithiation process is a soluble quinone species. Therefore, there is no formation of a solid residue or generation of additional gas, resulting in a minimal influence on the electrochemical performance.

### 3.1.2 | Prelithiation of anode materials

The anode capacity is higher than the cathode capacity in common battery designs. Hence, extra lithium will eventually be stored in the anode. Compared to prelithiating the cathode materials, prelithiating the anode materials is a more direct method. Various prelithiation strategies have been developed for different anode materials. It is a simple and effective way to achieve anode prelithiation by reducing the anode material with an organolithium reagent. Such an organolithium reagent must have high

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

lithium content, strong reductive capability, good stability, and removable byproducts after prelithiation. Li-Naph perfectly meets the above requirements, and thus, it is widely applied to prelithiate various anode materials. For example, Shen et al.[62] used Li-Naph to prelithiate a hard carbon (HC) anode material. Prelithiated hard carbon (pHC) was formed rapidly in a few minutes with electron and lithium ions being transferred from Li-Naph into the graphite layers of HC. As illustrated in Figure 4A, this prelithiation process was accompanied by the formation of a pre-SEI layer on the pHC electrode, which mainly had an organic composition (Li-OR). This pre-SEI layer could be converted into a dense and robust SEI film with the major components of inorganic LiF species when infiltrated by the electrolyte after initial discharge. Both pHC/$LiCoO_2$ and pHC/$LiFePO_4$ full cells tested in their study demonstrated high initial Coulombic efficiency (ICE) values (>95.0%) and nearly 100% utilization of the active electrode materials.

The Li-Naph reagent was also used for Si-based anode prelithiation. Yoshida et al.[65] used Li-Naph to prelithiate amorphous silicon particles. The rate and depth of the Li–Si alloy were easily controlled by adjusting the concentration of Li-Naph. From their research, a large initial irreversible capacity of the pristine Si electrode (1200 mAh $g^{-1}$) was successfully compensated after prelithiation in a 0.10 M Li-Naph complex solution for 1 h under these conditions.

Lithium biphenylide (Li-Bp) has oxidation–reduction properties similar to Li-Naph. Therefore, it can also be used as a prelithiation reagent. Yan et al.[63] reported a prelithiation strategy for $SiO_x$/C composite anodes by treating them with a Li–Bp solution. As demonstrated in Figure 4B, the homogeneity of $Li_xSiO_y$ was optimized by the combination of controllable complex solution prelithiation and a high-temperature calcination process, which could maintain the structural integrity of the LiBp-$SiO_x$/C anodes and reduce the irreversible consumption of Li ions. The pouch cell using $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ (NCM811) cathode and the as-prepared LiBp-$SiO_x$/C anode exhibited a high energy density (301 Wh $kg^{-1}$) and satisfactory cycling stability (93.3% capacity retention after 100 cycles). Therefore, the use of this complex solution is a worthwhile path to consider for prelithiating microparticles.

Li-9,9-dimethyl-9H-fluorene-tetrahydrofuran (Li-Flr) was also reported as an organolithium prelithiation reagent. The redox potentials of Li-Bp, Li-Naph, Li-Flr, and *n*-butyllithium (Li-n-butyl) are compared in Figure 4C, which was drawn by Zhang et al.[64] The redox potentials of these organolithium compounds increase in the order of Li-Flr < Li-Bp < Li-Nap < Li-*n*-butyl. Different redox potentials have a direct impact on the selection of prelithiation reagents and the prelithiation rate. Generally, a lower redox potential results in a shorter prelithiation time to reach a certain prelithiation depth. This means that Li-Flr has a faster prelithiation speed, which can shorten the prelithiation process and improve production efficiency.

Another category of anode prelithiation reagents is stabilized lithium metal powder (SLMP). Compared with organolithium reagents, metallic lithium has a higher lithiation efficiency and does not produce byproducts during prelithiation. However, metallic lithium powder is highly reactive. Therefore, it is necessary to build a

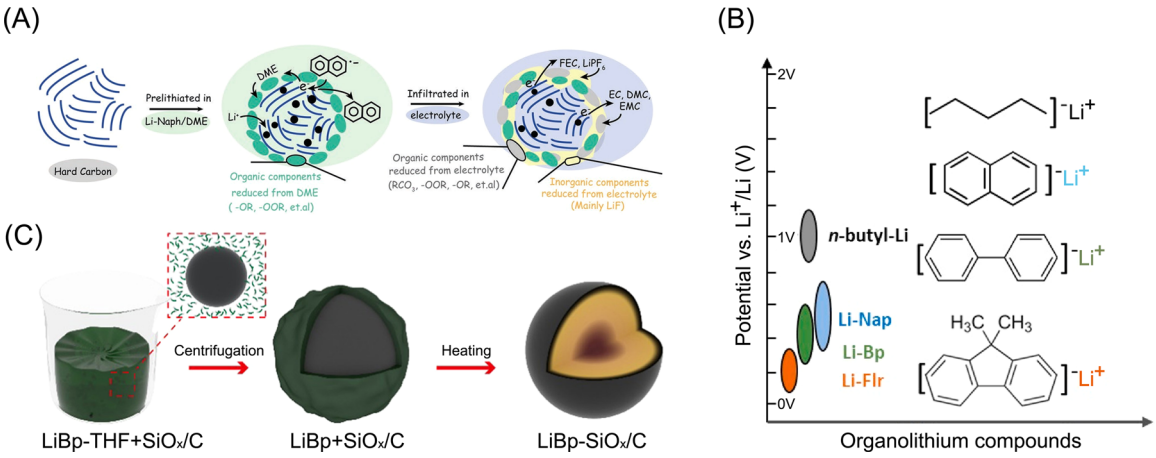



**FIGURE 4** Prelithiation of anode materials during their preparation: (A) schematic illustration showing the chemical prelithiation reaction of hard carbons (HC) in Li-Naph/dimethoxyethane (DME) solution and solid electrolyte interface formation on the prelithiated HC anode cycled in the Li-ion battery electrolyte. Reproduced with permission: Copyright 2020, Wiley.[62] (B) Schematic illustration showing the synthesis process of LiBp-$SiO_x$/C. Reproduced with permission: Copyright 2020, American Chemical Society.[63] (C) Redox potential and chemical structure of Li-Flr, Li-Bp, Li-Nap, and *n*-butyl-Li as prelithiation reagents. Reproduced with permission: Copyright 2020, Elsevier.[64]

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

passivation layer to stabilize lithium powder before mixing it with the anode material.

Jarvis et al.[66] prelithiated a carbon anode by simply mixing it with SLMP (produced by FMC Corporation) during the slurry preparation stage. In their test, the addition of only 1.93 wt% SLMP in the carbon increased the initial Coulombic efficiency from 77.9% to 95.4%. Moreover, SLMP is compatible with almost all kinds of anode materials, which makes it able to be widely used. Apart from common carbon anodes, it can be also used in solid-like electrolytes,[67] microsized SiO-based composite anode materials,[27] and lithiated poly(acrylic acid) aqueous binders,[68] showing a good prelithiation effect in all cases.

The benefits of SLMP are not limited to providing additional lithium sources. Wang et al.[69] found that SLMP could also effectively help SEI formation. With SLMP prelithiation, after resting at the open-circuit potential for a few days, both graphite half cells and graphite/NCM full cells showed a high first cycle capacity and Coulombic efficiency without a conventional formation process. The cell with SLMP without the formation process had even better cycling performance than the cell without SLMP but with the formation process. This meant that the cells with SLMP developed a fully functional SEI on graphite during the resting time. Hence, it is feasible to eliminate the conventional battery formation process by adding SLMP, significantly reducing the manufacturing cost of the cell.

However, the SLMP method is time-consuming due to the need for the electrolyte to fully react with the SLMP powder. To solve this problem, Forney et al.[23] proposed a pressure-activation strategy to assist SLMP prelithiation. From their experiments, an SLMP suspension was dropped onto the anode. After evaporating the solvents, the prelithiated SLMP anode was physically pressed in a compression mold. The contact area between the SLMP and Si surface was maximized by compressing the SLMP spheres onto the surface. This step ensured proper lithium diffusion into the anode and increased the consistency among the batteries in the present case. Their research showed that it was critical to properly apply the pressure-activation strategy to the SLMP during fabrication and that the coin cells must also be allowed to equilibrate for 40–50 h for the SLMP to properly diffuse into the anode under the current conditions.

Another anode prelithiation strategy is to physically use pure lithium metal to achieve prelithiation. Compared with the organolithium and SLMP methods, no reagents are used; thus, there is minimal impact on battery performance. For example, Zhao et al.[70] designed an Si prelithiation method in which $Li_xSi$ was synthesized by mechanically stirring a stoichiometric mixture (1:4.4) of Si nanoparticles (NPs, 50 nm in diameter) and Li metal foil at 200°C for 6 h in an inert atmosphere. Notably, trace oxygen in the environment helped to form a dense $Li_2O$ passivation layer on the outside of the $Li_xSi$ NPs, which prevented $Li_xSi$ from further being oxidized and enhanced the environmental stability of the prelithiated products. Furthermore, the $Li_xSi$ NPs reacted with the surfactant-like molecule (1-fluorodecane) dissolved in anhydrous cyclohexane for 2 h to form an artificial SEI layer.[71] Remarkably, the artificial SEI-protected $Li_xSi$ NPs showed a high prelithiation capacity of 2100 mA h $g^{-1}$ with negligible capacity decay in dry air after 5 days and maintained a high capacity of 1600 mAh $g^{-1}$ in the humid air.

A similar prelithiation method was applied to an Sn anode. Li et al.[72] prelithiated Sn by heating it with Li metal foil at 200°C while being magnetically stirred in a glove box. To stabilize the products, dispersed pyrrole monomers were oxidized and then precipitated on the surface of the Sn NPs, forming a conformal coating before prelithiation. The flexible polypyrrole (PPy) layer not only released the internal stress of $Li_xSn$/Sn induced by a large volume change but also prevented the active material $Li_xSn$/Sn from aggregating. Above all, with the protection of the PPy layer, the $Li_xSn$ alloy exhibited superior dry-air stability to fulfill the demand for practical battery fabrication.

Alaboina et al.[73] proposed another physical prelithiation method. In their experiments, the synthesized silicon nanoalloy (SiNA) material was first hand-milled with lithium stearate. The mixture was then placed into a furnace for prelithiation heat treatment. After heating at 300°C for 4 h, the lithium melted and diffused into the lithium stearate to fix the mechanical voids, and then the sample was cooled down to room temperature. The prelithiation products were hand-milled again for 10 min using a mortar and pestle. This facile prelithiation strategy helped to compensate for the Li loss, prevented parasitic capacity drain, and improved the volume expansion tolerance to a certain extent, as demonstrated by the long-term cycling test. The prelithiated SiNA demonstrated improved electrochemical performance, high Coulombic efficiency (~90% formation cycle), large capacity, enhanced cycle retention, less capacity fading, and low impedance, all of which were achieved at a significantly high loading (~2 mg $cm^{-2}$) for over 250 cycles.

### 3.1.3 | Prelithiation of binders and conductors

During the material preparation stage, the cathode/anode materials are mixed with a solvent, binder, and conductor to make a slurry. In addition to prelithiating

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

the cathode/anode materials, prelithiating binder is also a practicable method. Zhan et al.[74] proposed a binder-lithiating strategy for Si-anode batteries. Hard polyacrylic acid (PAA) and soft Nafion were partially lithiated by reacting with lithium hydroxide (LiOH) to form partially lithiated polyacrylic acid (P-LiPAA) and partially lithiated polynafion (P-LiNF), respectively. This partial lithiation process improved the transport of Li ions on the binder while retaining a large number of functional groups for subsequent reaction. Subsequently, the formed P-LiPAA and P-LiNF were mixed and stirred overnight at 40°C to form a hard/soft modulated trifunctional network (N-P-LiPN) via the hydrogen-binding effect. When applying this special prelithiation strategy to a silicon anode, the lithiated groups of N-P-LiPN could not only enhance the ionic conductivity but also provide an extra lithium source. Hence, the Si@N-P-LiPN electrode achieved a high initial Coulombic efficiency of 93.18% and stable cycling performance for 500 cycles at 0.2 C. Montree et al.[75] also reported a highly efficient binder prelithiation method. Prelithiated Nafion was synthesized via an ion-exchange process and was used as a binder for an Ni-rich cathode ($LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$, NCA). Their experiment showed that the initial Coulombic efficiency of the NCA cathode increased from 86% to 89.2% after using the prelithiated binder. Generally, the prelithiation of the binder is easy to achieve and compatible with traditional battery production lines. However, the binder material only accounts for a small proportion of a battery, which limits its maximum prelithiation capability.

## 3.2 | Prelithiation during the electrode processing stage

Prelithiation can also be conducted during the electrode processing stage. Unlike the prelithiation methods during the material preparation stage, prelithiation during the electrode processing stage usually requires new work steps and special equipment in the production line; thus, these strategies may increase the manufacturing cost. Nevertheless, the degree of prelithiation can be better controlled during the electrode processing stage. Therefore, prelithiation during the electrode processing stage is the most prevalent prelithiation method in the industry.

### 3.2.1 | Electrochemical prelithiation for electrodes

Electrochemical prelithiation is generally carried out by discharging or short-circuiting a temporary Li-metal half cell. It can achieve a quantitative and accurate lithium replenishment process and has been applied to various anode materials.

Silicon-based composites are widely considered the most suitable anode materials for prelithiation technology due to their high specific capacity and low ICE. Rezqita et al.[76] electrochemically lithiated Si/C electrodes at 0.1 and 0.5 V versus $Li^+/Li$ in a half-cell configuration. The full cell with Si/C prelithiated at 0.1 V exhibited a higher ICE in comparison to that with nonprelithiated Si/C (i.e., 86% vs. 25%). Kim et al.[77] introduced a scalable prelithiation scheme based on electrically shorting Li foil and successfully prelithiated an $SiO_x$ anode, as shown in Figure 5A. The degree of prelithiation could be accurately controlled by monitoring the voltage. This technique increased the ICE to 94.9% without sacrificing the structural stability of $SiO_x$ during cycling. Although their experiments were performed with coin cells, they provided a feasible design scheme on how to apply their method in a commercial battery production process, as shown in Figure 5B. Later, Lee et al.[79] prelithiated an $SiO_x$ electrode through a short circuit-containing constant-resistance (PLSC) process. By controlling the reaction time of the PLSC process, SEI layers successfully formed on the electrode, resulting in improved cycle life and power capability of the $SiO_x$ electrode.

In addition, direct contact of silicon-based anodes and Li metal foil in the presence of electrolytes can also realize prelithiation by internal short-circuiting. An Si nanowire (SiNW) anode was prelithiated when it was placed in direct contact with Li foil under pressure induction.[80] The degree of prelithiation could be controlled by adjusting the prelithiation time, and prelithiation for 20 min loaded ~50% of the full capacity into the SiNWs. However, it was difficult to exactly control the Li stoichiometry and SEI formation on the SiNWs in this way. To regulate the rate and degree of lithiation, a resistance buffer layer (RBL) was designed and inserted between the $SiO_x$ electrode and Li foil.[78] The porous structure, high electrical conductivity, and soft features of the RBL were beneficial to achieving homogeneous prelithiation of the $SiO_x$ anode (Figure 5C). The as-prepared lithiated $SiO_x$ anode exhibited a high ICE and specific capacity in both half cell and full cell.

The above prelithiation processes required the use of Li metal, which leads to additional safety hazards. Zhou et al.[81] prelithiated an Si electrode by using a new electrolytic cell without the use of Li metal. An Li-containing aqueous solution was utilized as both an aqueous electrolyte and an Li source. The assembled $MnO_x$/Si and S/Si full cells based on the prelithiated Si anodes showed high specific energies of 349 and 732 Wh $kg^{-1}$, respectively. It is noted that the specific energies cannot reflect cell-level data. However, the lithium

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

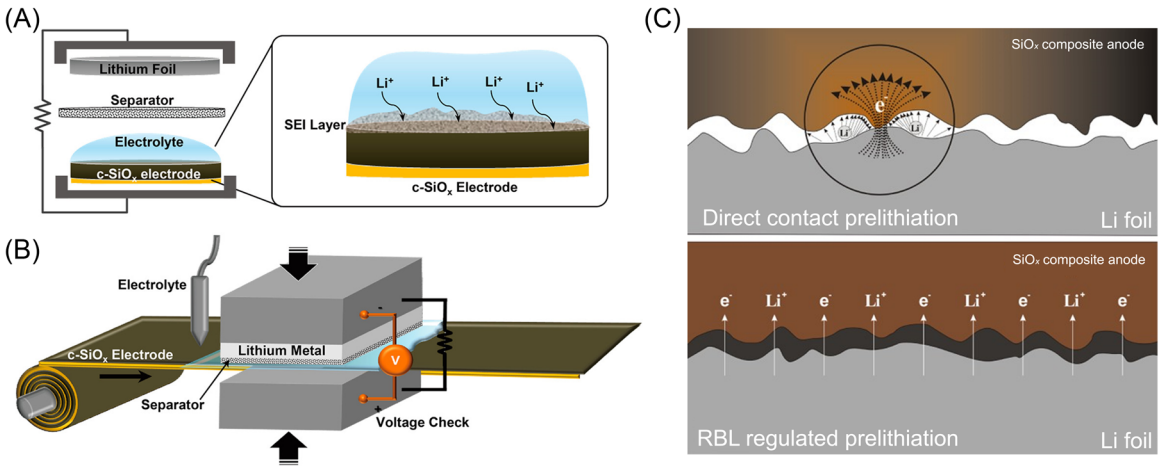



**FIGURE 5** Electrochemical prelithiation during electrode processing: (A) schematic showing the prelithiation process of the c-$SiO_x$ electrode; (B) scalable roll-to-roll process scheme of the prelithiation process of the c-$SiO_x$ electrode. Reproduced with permission: Copyright 2016, American Chemical Society.[77] (C) Schematic showing the comparison of the direct contact prelithiation process and resistance buffer layer-regulated prelithiation process. Reproduced with permission: Copyright 2019, American Chemical Society.[78] RBL, resistance buffer layer.

superionic conductor (LISICON, $Li_2O$–$Al_2O_3$–$SiO_2$–$P_2O_5$–-$TiO_2$–$GeO_2$) used in this electrolytic cell as a separator is expensive.

Sn is another anode material to store Li with a specific capacity of 994 mAh $g^{-1}$, which also suffers from an initial loss of active lithium during the SEI formation and the existence of some irreversible lithiation during the initial cycle. Hassoun et al.[82] prelithiated an Sn–C anode by utilizing the direct contact approach. Specifically, it was performed by placing the electrode in direct contact with an Li metal foil wetted by the electrolyte solution for 180 min. Notably, the irreversible capacity decreased from 63% to 14% at a specific current of 80 mA $g^{-1}$.

Similarly, the HC anode has a drawback with a low ICE of ~60% to 80% due to the large initial capacity loss.[83] Ezequiel et al.[84] prelithiated an HC anode by an electrochemical approach to investigate its influence on capacity. The prelithiation process involved galvanostatic charge–discharge cycles in half cells. With this strategy, the specific capacity of the $LiNi_{0.5}Mn_{0.5}O_2$/HC full cell increased from 92 to 162 mAh $g^{-1}$. Such a strategy has also been extended to improve the ICE of other anode materials, such as carbon nanospheres,[85] mesocarbon microbeads,[86] and PbS.[87]

The commercial graphite (Gr) has a relatively high ICE (typically >90%). Compared to other anode materials, prelithiated graphite is mainly used to improve the energy density and cycling stability of a battery. Sivakkumar et al.[88] achieved prelithiated graphite electrode by externally short-circuiting it with Li metal in Li/graphite/activated carbon 3-electrode cell, and the prelithiation capacity was estimated to be equivalent to 71% of the theoretical capacity of graphite after 10h. Decaux et al.[89] prelithiated graphite anodes by successive charge/self-discharge pulses in 2 mol $L^{-1}$ lithium bis(trifluoromethane)sulfonimide (LiTFSI) solution. Notably, the above two kinds of graphite prelithiation methods delivered higher energy densities than conventional methods.

Moreover, the electrochemical method can be adopted to obtain overlithiated cathodes. Wu et al.[90] studied how prelithiation affected the electrochemical properties of NCM materials with various chemical compositions. Notably, NCM materials that contained $Ni^{3+}$ could be easily activated by prelithiation at low voltages and accommodated the insertion of a double-layered Li substructure, thereby resulting in improvements in regard to the cycling stability and reversible capacity. Excess lithium was also introduced into an $LiMn_2O_4$ cathode to form $Li_{1.75}Mn_2O_4$ via an electrochemical method.[91] The as-formed $Li_{1.26}Mn_2O_4$/α-$Fe_2O_3$ nanorod cell delivered good cycling stability.

Overall, the prelithiation level can be accurately controlled with the electrochemical method. However, it usually involves troublesome disassembly of the half cells and reassembly of full cells, and the desired electrode may be slightly damaged during this process. In addition, the use of Li metal as an Li donor poses safety hazards during the fabrication process. The complexity and potential safety hazards limit these strategies to be only applicable in laboratory research. Thus, they have a long way to go before being practically applied.

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

### 3.2.2 | Chemical prelithiation of electrodes

Chemical prelithiation is achieved by immersing the electrode into reductive lithium-containing solutions, which prevents direct contact with metallic lithium to increase safety and stability. Generally, as illustrated in Figure 6A,[92] the Li-containing solution is composed of a strongly reductive prelithiation reagent and chemically stable solvent. When the Li-containing solution meets the electrode material, the active lithium spontaneously transfers into the electrode material due to a chemical potential difference.

As a typical prelithiation reagent, Li-Naph has been used in various electrodes due to its mild reactivity and active lithium source. By a spontaneous chemical reaction with Li-Naph, HC can be quickly and controllably prelithiated in a few minutes, and a desirable prelithiation level can be achieved by adjusting the reaction time.[62] The prelithiated HC anode has a dense and robust SEI layer consisting of uniformly distributed LiF, which exhibits a high ICE of ~99.5%. Shen et al.[94] used Li–Naph to fully prelithiate sulfurized polyacrylonitrile (S-PAN) composite into an $Li_2S$-PAN cathode and partially prelithiate nanosilicon into an $Li_xSi$ anode to eliminate its initial irreversible capacity loss. The Li-Naph reagent is ideally suitable for partially prelithiating Si-based anodes since its redox potential (0.35 V) is just below the SEI formation potential (~0.5 V) and slightly higher than the $Li^+$ insertion potential of Si. After prelithiation, the Si/$Li_2S$ cell demonstrated a nearly 100% capacity utilization of both the cathode and anode materials, achieving high specific energy of 710 Wh $kg^{-1}$.

Compared with Li-Naph, Li-biphenyl (Li-Biph) has similar chemical properties but stronger reducibility, making it a good alternative to prelithiation reagent. Zhang et al. used an Li-Biph/THF solvent to prelithiate an HC anode (Figure 6B).[93] When matched with an $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ cathode, the full cell with the prelithiated HC anode exhibited an improved ICE (90.2% vs. 75%) and better cycling performance (93.4% vs. 74% after 200 cycles at 1 C) compared to those of the pristine full cell. Additionally, the reagent was also used to prelithiate P, Sn, P/C, and $SnO_2$/C anodes, and compared to the above improvement, a much higher ICE and even better cycling performance were realized.[92,95,96] In addition, a cathode prelithiation method using $Li_2S$-PAN as a lithium "donor" was synthesized via a chemical reaction between S-PAN and Li–Biph complex. When $Li_2S$-PAN was loaded on an $LiFePO_4$ electrode, the $LiFePO_4$/$Li_2S$-PAN composite cathode displayed a higher initial charge capacity than that of pristine $LiFePO_4$.[97]

However, both Li-Bp and Li-Naph with a redox potential of >0.3V (vs. $Li^+$/Li) exhibit a low reducing ability. To solve this problem, Shen et al.[98] found that 2-methyl tetrahydrofuran (2-Me-THF) solvent with more effective electron donation ability could negatively shift the redox potential of Li–Biph. The obtained Li–Biph/2-Me-THF solution showed an extremely low redox potential (<0.1 V vs. $Li^+$/Li), which could controllably prelithiate the Gr anode to the desired level and exclude solvent cointercalation into the Gr anode. All the full cells with the prelithiated Gr anodes exhibit improved ICEs and energy densities. Additionally, the Li-Biph/2-Me-THF solution was applied to the blended Gr/Si

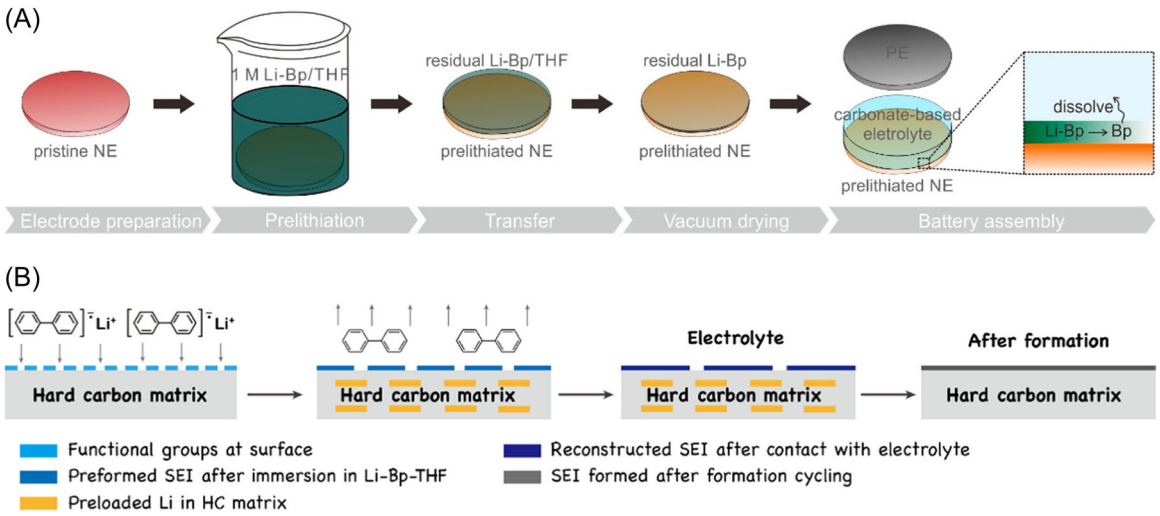



FIGURE 6 Chemical prelithiation during anode processing. (A) Schematic illustration of the prelithiation procedure in ambient air. Reproduced with permission: Copyright 2019, American Chemical Society.[92] (B) Proposed prelithiation mechanism of Li-Bp-THF in HC. Reproduced with permission: Copyright 2020, American Chemical Society.[93] HC, hard carbon; THF, tetrahydrofuran.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

systems to improve their ICEs. Given the high ICE (~100%) of the prelithiated Gr/$SiO_x$ composite anode, the full cell exhibited an energy density of 506 Wh $kg^{-1}$ (98.6% of the ideal value).[99]

Furthermore, Li-containing reagents with low redox potential have been prepared to effectively lithiate electrodes with low lithiation potential. Jang et al.[100] developed Li-Biph derivatives with low (<0.2 V vs. $Li^+$/Li) redox potential, and these could cause $SiO_x$ anodes to accommodate active Li, thereby elevating the ICE to exceed 100%. Recently, a new reagent, Li–9,9-dimethyl-9H-fluorene/THF, was shown to have a couple of Flr/$Flr^-$ redox peaks at 0.07 and 0.3 V (vs. $Li/Li^+$). Due to its molecular structure, it has a lower potential than Biph and Naph, which makes the electrons and Li ions easily transferred to reduce the SiO/Gr anode; thus, fast prelithiation of the SiO/Gr anode and an improved ICE could be achieved.[64] A high ICE of approximately 90.7% in the half cell was attained after 10 min of prelithiation. When paired with the $LiNi_{0.6}Co_{0.2}Mn_{0.2}O_2$ (NCM622) cathode, the ICE of the full cell with the prelithiated graphene-coated SiO (SiO/Gr) anode was increased to 87.1% compared with 61.1% of the full cell containing the cathode without prelithiation.

In conclusion, chemical prelithiation is a simple and universal method to improve various electrodes. Benefiting from the strong lithiation ability of prelithiation solutions, the desired electrode can generally be rapidly prelithiated in a few minutes, and the prelithiation level can be tuned by the reaction time. Such a prelithiation method has been used on a small scale in some battery industries. However, it should be noted that both the prelithiation reagent and prelithiated electrode are unstable in air. Thus, special cleaning and drying steps in an inert atmosphere are needed in the manufacturing process, which may increase the difficulty of industrial application. Moreover, the synthesis of prelithiation reagents is not facile due to strict condition requirements, and the low specific capacity (<400 mAh $g^{-1}$) of the reagents is not conducive to the industrialization of high-performance LIBs. Therefore, developing prelithiation reagents with high lithiation activity, good safety, and electrode compatibility is still necessary and challenging.

### 3.2.3 | Mechanical prelithiation for electrodes

Mechanical prelithiation usually refers to adding active Li onto electrodes through mechanical pressing without the use of an electrolyte, which simplifies the operational procedure, improves safety, and reduces the cost. Under these circumstances, an Li metal foil with proper thickness and appropriate mechanical property matters. Nevertheless, pure Li metal foils that are commercially available are usually too thick (typically 50–750 μm) or too mechanically fragile for mechanical prelithiation. However, Chen et al.[101] synthesized an ultrathin (0.5–20 μm) mechanically robust Li@eGF (expanded and reduced graphene oxide film) film by infiltrating molten lithium into ultrathin graphene oxide. By covering a 2-μm-thick Li@eGF film (0.391 mAh $cm^{-2}$) on a graphite anode, the ICE of the graphite||Li half cell increased from 93% to approximately 100%, and 5-μm-thick Li@eGF (0.853 mAh $cm^{-2}$) could precisely prelithiate a nanosilicon anode and increase the ICE of the anode from 79.4% to 100.5%.[101] Moreover, a self-supporting Sn foil is a promising anode material due to its high volumetric capacity and mechanical properties; unfortunately, it suffers from a shockingly low ICE (10%–20%). However, the reaction between the Sn anode and Li metal during the mechanical prelithiation process will significantly improve the ICE of the Sn anode. Xu et al.[102] developed a roll-to-roll mechanical approach of metallurgically prealloying Sn with Li to form a highly uniform $Li_xSn$ foil. Specifically, stacked Sn and Li foils were pressed by a roller, and an $Li_xSn$ layer on the surface of the Sn foil was produced by the spontaneous reaction at the interphase between the Sn and Li metal (Figure 7A). As a result, two adjacent layers of $Li_xSn$ and Sn substrate could be identified, and the thickness of the $Li_xSn$ layer was fairly homogeneous (Figure 7B). The assembled $LiFePO_4/Li_xSn$ full cells showed an increased ICE (from 20% to 94%) and achieved 200 stable cycles at ~2.65 mAh $cm^{-2}$. The air stability of the $Li_xSn$ foil made it easy to handle in an open-air environment. Later, a more uniform prelithiation and thinner Li-containing foil anode was achieved by grain size refinement, which was performed by doping with Ag and Cu.[103] The full cells pairing the as-prepared thinner $Li_xAg_3Cu_{0.5}Sn_{96.5}$ foil against both commercial $LiFePO_4$ and NCM523 cathodes exhibited satisfying battery performance. This roll-to-roll mechanical method was also utilized to prepare lithiated Al-based anodes with good electrochemical cycling performance.[104]

Even though it is difficult, researchers found that lithium could intercalate into graphite by directly pressing lithium metal into graphite. This greatly expands the application field of the roll-to-roll mechanical method, as graphite is still the most commonly used anode material for commercial LIBs. This method could have the lowest cost since it only involves physically utilizing lithium metal without using any other chemical reagents or energy-intensive processes. However, only a few contacted layers of lithium can successfully intercalate into graphite. To improve the utilization efficiency of lithium and reduce the formation of dead lithium, extremely thin lithium foil is required in this method. This results in a large engineering problem because thin lithium foil does not

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

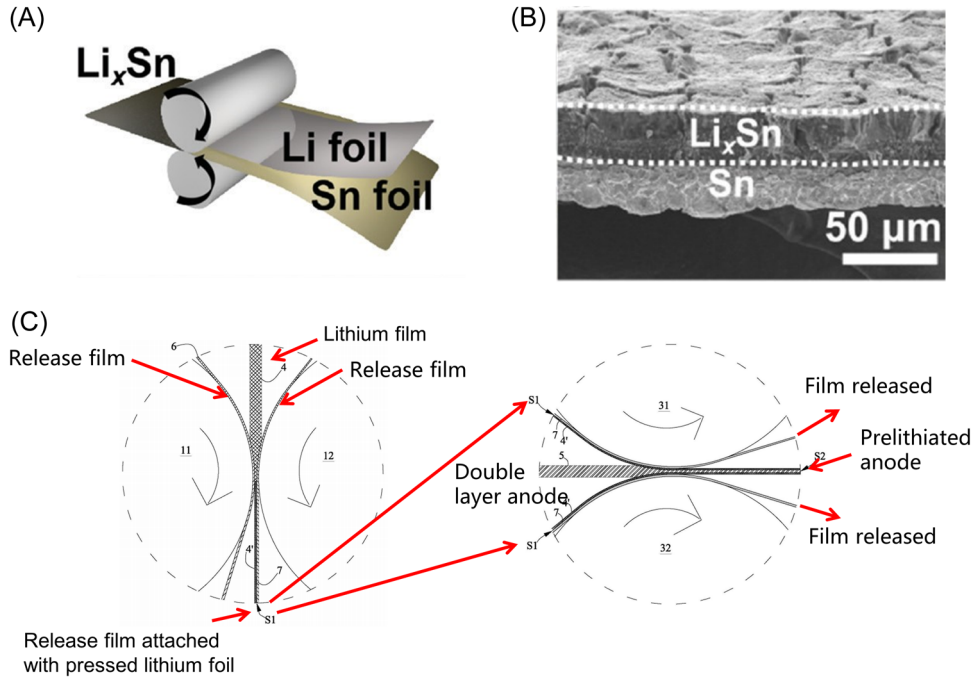



**FIGURE 7** Mechanical prelithiation during electrode processing: (A) schematic diagram of the roll-to-roll prelithiation method for preparing electrodes; (B) $Li_xSn$ layer formed on an Sn substrate without the introduction of other residues. Reproduced with permission: Copyright 2019, The Royal Society of Chemistry.[102] (C) Schematic diagram of the roll-to-roll prelithiation method with a graphite anode.

have enough strength, and the preparation and application conditions of thin lithium foil are too strict. To solve this problem, Ningde Contemporary Amperex Technology Limited[43,105,106] invented a novel method based on polymer release films. As shown in Figure 7C, a thick lithium film is pressed with a polymer release film to form a complex film with an extremely thin lithium layer on the surface. The release film provides enough strength to manipulate the complex film. During prelithiation, the complex film is pressed onto the anode, and then the release film is peeled off to realize lithium transfer. This method achieves high lithium utilization efficiency, controllable lithiation degree, high lithiation uniformity, and good industrial feasibility. Thus, it has already been applied to fabricate commercial prelithiated LIBs on a small scale.

In short, mechanical prelithiation has already shown promise toward scalable applications because the prelithiated electrodes are relatively stable in an ambient air environment. However, it is still a great challenge to precisely control the prelithiation degree. The addition of lithium in excess of the anode capacity may lead to harmful lithium dendrites. Ultrathin and free-standing Li metal foils (≤20 μm) are attractive for precise prelithiation, but facile fabrication approaches with high safety are lacking. Given that the introduction of other possible residues is prevented, the generalization of this method to other forms of electrodes is highly desirable.

### 3.2.4 | Noncontact prelithiation methods for electrodes

Noncontact prelithiation methods, such as evaporation, have been investigated to accurately control the thickness of the deposited lithium and improve the utilization of the Li. The most advanced theory proposed by Zhang et al.[107] shows that the initial contact state of the Li source/anode interface has been significantly promoted by depositing an Li film onto the anode by vacuum thermal evaporation (VE) method other than the traditional mechanical rolling (MR) method (Figure 8). The Li source/anode interface of the VE-anode (the anode obtained by the VE method) contains a higher electron channel density than the MR-anode (the anode obtained by the MR method), which could accelerate the Li-metal dissolution. Due to its film formation mechanism, the VE-anode owns rapid reaction kinetics, more homogeneous current density, more uniform SEI film thickness, and higher Li utilization. Moreover, the Li film dose can be monitored by a crystal oscillator and the amount of Li film can be well regulated by the VE method. Due to the high efficiency of the vacuum thermal evaporation method, the half cells with VE-anode exhibit higher initial Coulombic efficiency at a 0.1 C rate, and lower activation energy of the Li-ion transport through SEI and charge transfer. The VE-4 (depositing 4.0 μm Li film onto the anode by the VE

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

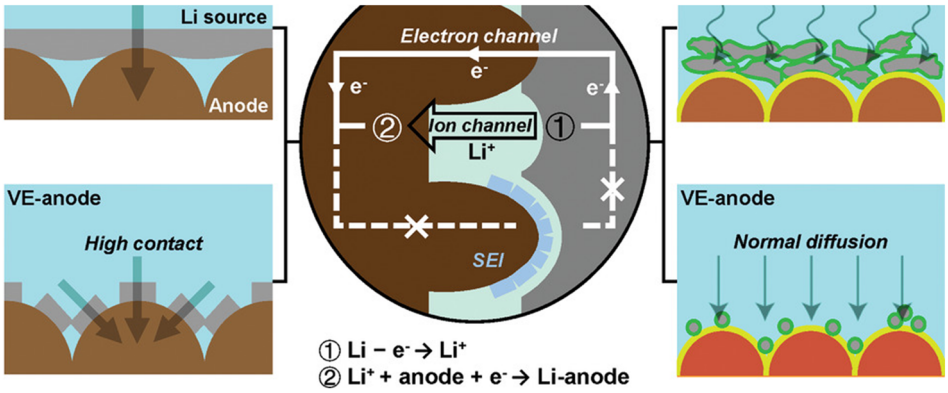



FIGURE 8 Schematic of the null MR- and VE-guided prelithiation from the perspective of Li source/anode interface evolution. Reproduced with permission: Copyright 2022, Wiley.[107] MR, mechanical rolling; VE, vacuum thermal evaporation.

method) has achieved an Li utilization rate of 91.0%, while the MR-4 (attaching 4.0 μm Li film onto the anode by the MR method) achieved only 74%. The pouch cell, which was assembled by using an $LiNi_{0.8}Co_{0.1}Mn_{0.1}O_2$ and a VE-4 anode under an N/P ratio of 1.05, presented capacity retention of 95.8% after 600 cycles at a 0.2 C rate. Conceptually similar work has also been carried out by Kim et al.[108] in which a precisely prelithiated Si-Gr anode was prepared by a vacuum thermal evaporation method when utilizing a certain evaporation rate of lithium metal. The scanning electron microscopy (SEM) images showed that the dark gray surface of the anode became white after Li deposition, confirming the presence of deposited Li. After cell assembly, the Si-Gr electrode was spontaneously lithiated by the deposited Li, forming a prelithiated Si-Gr electrode. The amount of deposited Li affected the initial open-circuit voltage (OCV) and the ICE of the electrode, and 10% prelithiated Si-Gr was selected for the subsequent full-cell experiment. Regarding the $LiCoO_2$/Si-Gr full cell, the ICE increased from 76.4% to 92.5% after prelithiation. A similar method was also reported to prelithiate $SiO_x$-based anodes.[109]

Recently, Yang et al.[110] reported a strategy of prelithiation by fixing metal lithium and in situ synthesizing α-$Li_3N$ film on the cathode surface. Lithium evaporation was accomplished first, and the in situ conversion of metal lithium to crystallized lithium nitride was subsequently performed. Except for breaking the match barrier between $Li_3N$ and the polar solvents, this approach was inexpensive, easy to perform, and provided precise control. The synthesized α-$Li_3N$ film compensated for the lithium loss in the $LiFePO_4$/graphite full cell. In addition, a prelithiation method based on chemical vapor deposition (CVD) was reported to prelithiate the electrode materials of LIBs.[111] The devices for CVD prelithiation include cathode and anode material pretreatment devices, lithium source sublimation or cracking devices, deposition chambers, and graded discharge devices. The lithium source is atomized in the CVD process, evenly coated on the surface of the cathode and anode materials, and diffused to the inside of the materials, exhibiting strong adhesion and good stability.

In short, the precise controllability of these methods makes them appealing for commercial use. However, there is still much space to further extend them to other electrode materials.

## 3.3 | Prelithiation during the cell assembly

Theoretically, since cell assembly is the fabrication step after material preparation and electrode processing, conducting prelithiation in this step does not require changing the equipment for the previous cell fabrication. Due to this industrial convenience, this method has attracted much interest from many battery developers. However, at present, there are still many technical challenges.

Extra lithium can be added by involving a sacrificial lithium metal electrode in the cell, as shown in Figure 9A. With the device designed by Huawei,[47] the prelithiation degree could be precisely controlled by the discharge current and discharge time. Furthermore, this prelithiation method could be implemented not only during manufacturing but also battery working process.

Regarding prelithiation in the ultimate part of the fabrication process of LIBs, more consideration needs to be given to industrial convenience. One of the most important things is to ensure that lithium ions can transfer between different electrode layers. Tatsuya et al.[112] proposed an optimization method in 2019, as illustrated in Figure 9B. By perpendicularly punching electrodes and collectors, direct Li-ion diffusion pathways in the cell were created. The first anode close to the lithium strip was

26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTRECPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

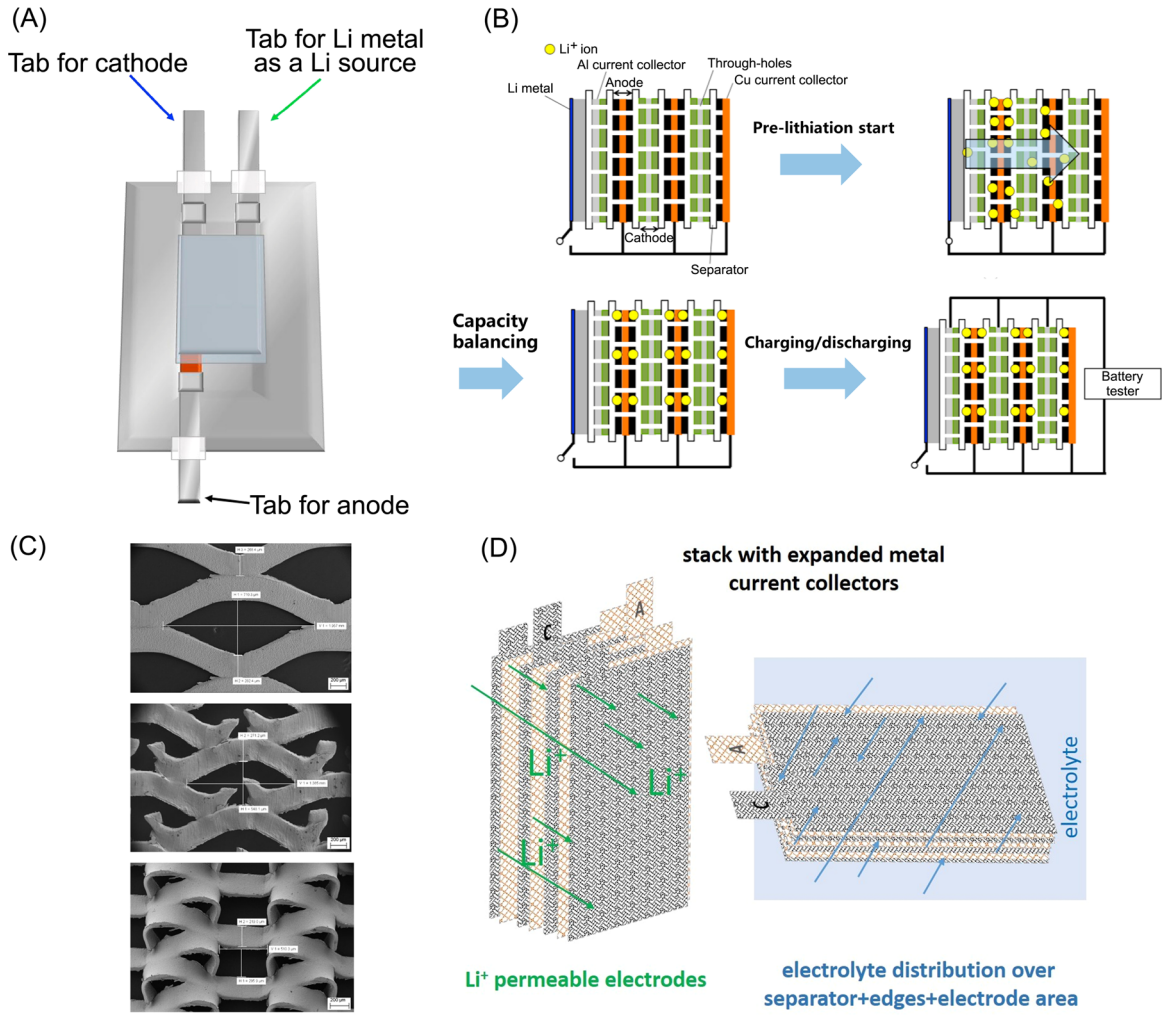



**FIGURE 9** Battery prelithiation during the assembly stage: (A) schematic description of the laminated cell with a special lithium metal electrode; (B) schematic of a series of perpendicular electrodes and collectors undergoing predoping, capacity balancing, and charge/discharge processes in a laminated cell composed of porous anodes and cathodes. Reproduced with permission: Copyright 2019, Pergamon Press.[112] (C) Scanning electron microscopy image of the expanded metal collectors for a lithium metal electrode cell; (D) schematic of the lithium ion and electrolyte transfer path in a lithium metal electrode cell. Reproduced with permission: Copyright 2020, Ulm University.[113]

efficiently prelithiated, while other anodes also exhibited decent effects owing to the dispersion of straight passageways. Notably, after prelithiation, a period of time was needed to achieve capacity balancing, which was actuated by charge compensation. Next, by comparing various punched hole diameters and capacity balancing times, the optimized solution was established. The optimal diameter of 20 μm and capacity balancing time of 12 h resulted in the increased ICE to over 97% and almost 100%, respectively, after the second cycle. In addition, with an optimized prelithiated charge amount, the ICE further improved to nearly 99%. With a similar mindset, Zensen et al.[113] also reported a prelithiated cell structure. Different from punching holes in the collectors, expanded metal collectors with three-dimensional pores (Figure 9C) were used to manufacture cells with lithium metal electrodes. As demonstrated in Figure 9D, the lithium ions and electrolytes could freely travel between the different electrodes. An $LiFePO_4$(LFP)/Si tri-electrode pouch cell with this innovative current collector structure was stabilized for up to 150–100 cycles by prelithiating the Si anode to 5%, 10%, or 15% SOC. By increasing the prelithiation degree to an exemplary 40% SOC, the stability of more than 400 cycles was achieved. Due to the admirable practicability of the method, it could be applied in large-scale battery fabrication processes.

## 4 | PROBLEMS CAUSED BY PRELITHIATION TECHNOLOGY

Although prelithiation brings a lot of advantages for LIBs, it may also cause many problems in regard to the design, fabrication, operation, and recycling. As shown in

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Figure 10, these problems can be sorted into several levels, including materials, electrodes, cells, and manufacturing-related ones. In the following section, we will discuss some typical problems caused by prelithiation, as well as possible solutions.

## 4.1 | Residual lithiation reagent/additive-induced parasitic reactions

In most prelithiation methods, different reagents/additives are used to help the formation of lithiated anode materials, overlithiated cathode materials, or lithium-rich products. There is considerable evidence that even when a washing and drying process is used, some of the reagents/additives still remain behind. Usually, the reagents/additives have high chemical activity and can easily cause side reactions, such as oxidation, decomposition, or gas release, and hence adversely affect battery performance.

For instance, THF[63,93] and tetrahydropyran (THP)[99] are commonly used in the prelithiation process. Ideally, these reagents should be removed by a low-temperature evaporation process. However, such a postprocessing method cannot eliminate all of the solvents; thus, there is always a trace amount of solvent adsorbed on the surface of the electrode material particles. This residual solvent may participate in side reactions in the battery, consuming reactive lithium ions and increasing the internal resistance of the battery.

Sacrificial additives used for prelithiation can also cause negative effects on battery performance. Common sacrificial additives, such as $Li_3N$,[55,114] $Li_2O$,[115] $Li_2C_4O_4$, $Li_2C_3O_5$, and $Li_2C_4O_6$, decompose and release gases (e.g., $N_2$, $O_2$, etc.) in the first charge process, which damages the structure of the electrode and blocks the transport of Li ions.[57] In contrast to the above sacrificial additives, Li metal oxides, such as $Li_2NiO_2$,[58,116] $Li_5FeO_4$,[59,117] $Li_2CuO_2$,[118] and $Li_6CoO_4$,[56,119] do not generate gases. However, after releasing active Li ions, these compounds leave behind solid residues with poor conductivity, which increases the battery resistance and decreases the energy density.

In conclusion, the current commercialized LIBs have been developed and perfected for years. Prelithiation technology inevitably involves the intentional introduction of new compounds into battery systems. The impact of residual reagents/additives after the prelithiation process should be carefully examined, and an extra washing step may be needed after prelithiation.

## 4.2 | Changes in the SEI formation conditions

The quality of SEI is one of the most important issues regarding achieving highly reversible cycling for LIBs. To guarantee a high-quality SEI, the applied current and voltage in the formation process need to be carefully optimized to control the SEI growth rate.

However, for prelithiated LIBs, the SEI formation process differs significantly from that for conventional LIBs. In the prelithiation process, the passivating reactions between the anode and electrolyte are completed earlier, and the state of the anode particle surfaces is different from that of an unprelithiated anode. Once electrochemical activation is triggered, the surface will suffer from a change in the SEI composition, resulting in variations in chemical stability and mechanical properties, which may lead to the degradation of battery performance. It is worth evaluating the cycling stability of prelithiated batteries with various interfaces due to electrochemical activation. Special pretreatments are needed to guarantee the quality of the SEI. Some pioneer works have proposed various strategies to enhance the SEI stability during the prelithiation process. For example, Shen et al.[62] developed a chemical prelithiation

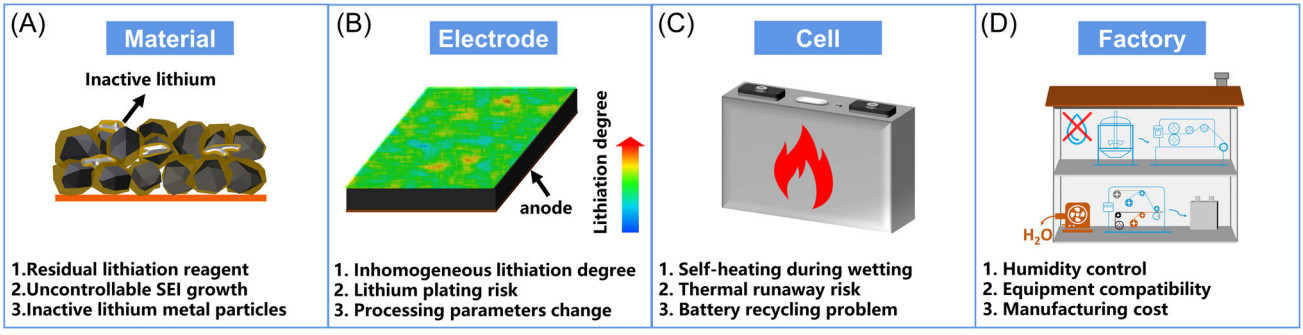



FIGURE 10 Problems caused by prelithiation technology: (A) material level; (B) electrode level; (C) cell level; and (D) manufacturing-related issues.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

method to improve the ICE of HC. They found that the SEI formed during prelithiation was thinner, denser, and more robust than the conventional SEI formed during the first charge. However, it should be noted that such an ex-situ method may damage the SEI film and result in imperfection during the subsequent cleaning, drying, rolling, and cell assembly steps. Therefore, it is necessary to develop postassembly SEI repair technology to enhance the quality of the final SEI in the full cell.

The SEI formation process is even more uncontrollable for prelithiation methods using lithium metal as the lithium source, such as the mechanical rolling and SLMP methods. From the aspect of the prelithiation driving force, Li-ions intercalating in the anode spontaneously occurs when the electrolyte is injected into the battery. As shown in Figure 11, with the unrestrained extent of the reaction, the spontaneous intercalating reaction releases a large amount of heat. The temperature of the cell quickly increases and significantly influences the formation process of the SEI. The maximum temperature of this exothermic process is primarily determined by the structure of the battery, electrolyte injection rate, and surface area/passivation quality of lithium metal. Using highly passivated SLMP may alleviate the heating rate, prolong the SEI formation period, and reinforce the quality of the SEI[69]; however, the possibility of the initial formation of unfavorable dead lithium remains.

## 4.3 | Potential for increased lithium plating

Lithium plating is a common undesirable phenomenon on the anode side, which generally occurs with lithium intercalation during the charging process.[120–124] In many cases, prelithiation may increase the possibility of lithium plating for the following reasons.

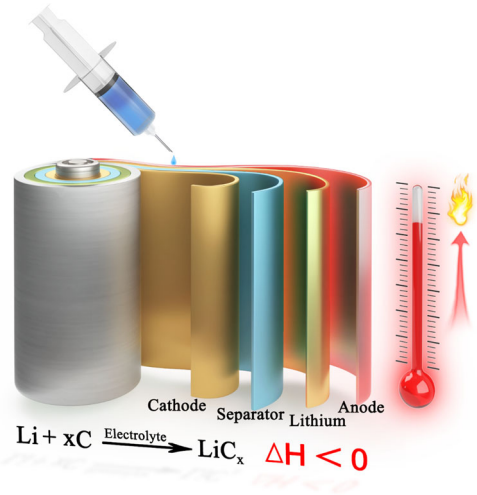



FIGURE 11 Self-heating problem during the electrolyte wetting process of a prelithiated lithium-ion battery with a lithium metal component.

Normally, lithium ions insert into the vacant sites in the interlayer spaces of graphite[125] in the potential range of 65–200 mV versus $Li^+/Li$.[126] However, when the anode potential is below 0 V versus $Li^+/Li$, lithium-plating is thermodynamically permitted.[127,128] Under this principle, if the Li intercalation rate in the anode is higher than the solid-phase diffusion rate of Li ions in the anode, Li ions are prone to accumulate on the anode surface, possibly nucleate, and subsequently grow into Li dendrites. As a common perception, prelithiation treatment lowers the average anode potential during charge/discharge, which brings a high risk of lithium plating.

From the dynamic perspective during the charging process, the solid-phase diffusion rate of lithium ions in the anode is slow when the battery is near the fully charged state, the energy barrier of the leap of the lithiated graphite phase transition from $LiC_{12}$ to $LiC_6$ is much higher, and the driving force required is larger, resulting in a more unfavorable Li-ion intercalation in the anode.[129] Moreover, due to the lower solid-phase diffusion rate and the effect of polarization, lithium dendrites will occur when the sum of the equilibrium potential and overpotential is below 0 V versus $Li^+/Li$ during charging. Furthermore, at the end of discharge, because of prelithiation, the anode has a lower surface potential than that without prelithiation. Therefore, lithium dendrites appear in a minor overpotential condition.

The nonuniformity of the lithiation degree is a common problem for most prelithiation techniques. This may cause uneven lithium insertion resistance and current density, which further increases the possibility of local lithium plating. Lithium dendrite growth is usually accompanied by fresh SEI formation and gas release. The amount of gas is usually very low, but it is still possible to block the migrating path of Li ions, resulting in a more severely uneven distribution of charge current and aggravating lithium plating. Such a self-accelerating process may cause a remarkable loss of active lithium and capacity decay.

## 4.4 | Thermal runaway risk

The thermal runaway of LIBs is widely considered the most important obstacle to their further popularization. Prelithiation may increase the risk of battery thermal runaway for the following three main reasons.

First, as described in the previous section, the prelithiation process usually results in a high lithium

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

plating possibility. In most studies, the lithium-plating dendrites, which overgrow and pierce the separator to cause an internal short circuit, are often considered the main cause of thermal runaway. The high current of the internal short circuit will create a transient high temperature. The thermal runaway process is activated once the local temperature exceeds the decomposition temperature of the SEI.

Second, in the prelithiation process, it is difficult to guarantee that 100% of the added lithium can be turned into active lithium ions and inserted into the anode in the first cycle. Hence, there is a huge chance that the prelithiation process leads to plenty of dead lithium inside the battery. This part of the dead lithium will not impact the battery's working performance during ordinary charge/discharge. However, Feng et al.[130] found that these lithium particles would severely react with the electrolyte and accelerate thermal runaway if the battery is overheated. In their research, the main source of heat for thermal runaway was the decomposition of the SEI film, and the main source of heat generation came from the redox reaction between the cathode and anode electrodes. Due to the decomposition of the SEI film occurring at a relatively low temperature of 69°C,[131] these residual lithium molecules on the electrode surface could reduce the stability of the SEI during the cycling by reacting with the electrolyte, thereby releasing a large amount of gas and Joule heat. For comparison, the internal short circuit is generally considered a primary contributor to heat generation but actually only accounts for a fraction of the total heat generated in the thermal runaway process. In regard to prelithiated batteries, the additive lithium metal may induce a new thermal runaway path because of the high reducibility of lithium metal. In the case of the redox reactions caused by lithium metal, the regulation and control strategies for the thermal runaway of prelithiated batteries need to be taken seriously.

Third, prelithiation increases the energy density of the battery. The anode material of prelithiated batteries always has a higher lithium insertion state than normal materials. The in situ high-energy X-ray diffraction (HEXRD) analysis in Figure 12 shows the different decomposition temperatures of the lithiated graphite with electrolyte.[132] Unsurprisingly, a higher lithiation degree corresponds to a lower decomposition temperature. It is well known that the prelithiation process can improve the overall lithiation degree of the anode, which eventually lowers the thermal stability of the battery.

## 4.5 | Challenges of battery management

In application scenarios, a battery management system (BMS) normally plays an important role in regulating

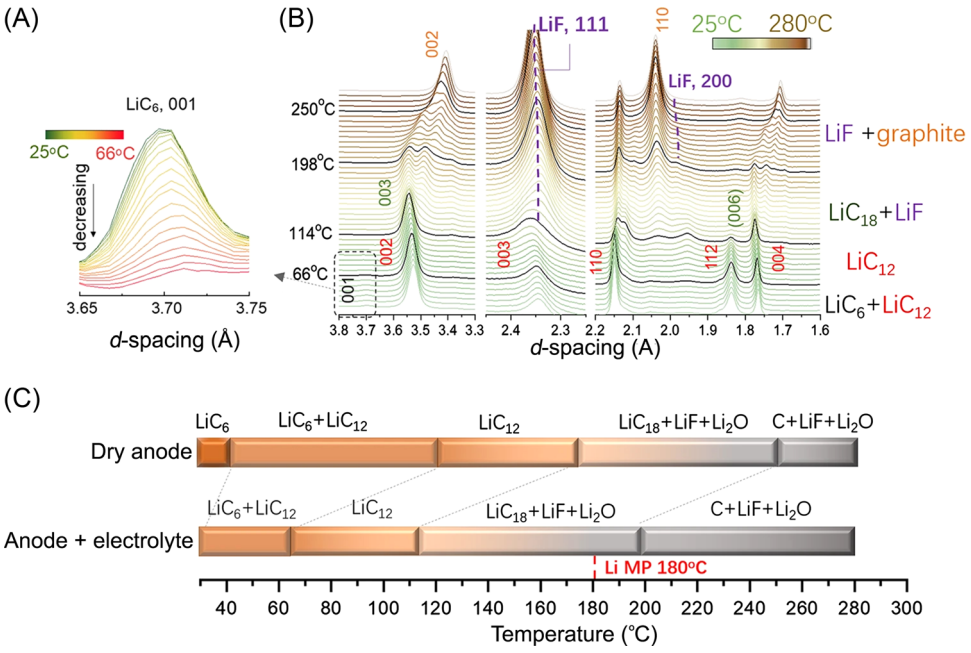



FIGURE 12 Structural evolution of lithiated graphite with the electrolyte during heating: (A) HEXRD plots of the $LiC_6$ (001) Bragg reflection evolution from 25°C to 66°C; (B) HEXRD plots of lithiated graphite anode with the electrolyte from 25°C to ~280°C at a rate of 2°C/min. The 2θ value was converted into D-spacing following Bragg's rule; (C) comparison of the phase evolution of the lithiated graphite anode with and without the presence of an electrolyte. The lithium melting point (MP) at 180°C is also described in the figure. Reproduced with permission: Copyright 2021, Nature.[132] HEXRD, high-energy X-ray diffraction.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

and optimally controlling battery modules. From the perspective of BMS, one of the key issues is to estimate the SOC and state of health (SOH) of every single battery accurately, for the sake of ensuring the consistency and balance between individual cells in the battery module. The estimation accuracy of the two indexes is critical to battery safety and is still a great challenge for conventional battery operation.

When referring to prelithiated LIBs, such problems are prone to be magnified. Prelithiation changes the SOC variation window of the anode, resulting in a different voltage interval. Meanwhile, the addition of extra lithium also changes the power and impedance characteristics of the battery. Therefore, traditional SOC estimation algorithms based on electrochemical property data, such as OCV[133,134] and impedance change,[135,136] are no longer applicable. Among all the electrical data of LIBs, the most critical and relevant data for SOC estimation is the OCV. However, the SOC and voltage relationship will change nonlinearly due to the different battery systems or operating conditions. This means that it is difficult to use the same model to perform accurate SOC estimation on various practical batteries or other situations. Moreover, the lithiated battery has a longer and flatter voltage "plateau" than the conventional battery, which means that the voltage change is more unapparent during operation.[137] Different lithiation degrees of the battery will affect the relative shift of the cathode and anode potential intervals. The plateau period of the output voltage of the battery will also change during use, which will further increase the difficulty of SOC estimation accuracy.

Compared to SOC estimation, SOH prediction is more complicated. SOH is commonly used to predict the remaining life of a battery. Hence, it is generally defined by the end-of-life criterion as the residual capacity dropping to approximately 80% of the initial capacity or the internal resistance increasing to approximately twice the initial state. Accurate prediction of the remaining available life of LIBs under different operational conditions can not only ensure safe and reliable operation but also optimize the utilization of the remaining value of the used batteries.[138] Capacity degradation is usually associated with intrinsic and extrinsic factors of the battery, such as degradation reactions inside the battery and various outside abuse. The resistance increase is generally associated with side reaction effects, such as SEI film growth, gas generation, and electrolyte deterioration. The traditional BMS often predicts the SOH based on the battery aging model driven by a mass of battery decay data.[139]

However, prelithiation changes the aging mechanism of the battery. Excess lithium keeps the anode material in a high-energy state, enhancing the side reaction on the anode surface, changing the SEI growth speed, and even changing the side reaction species. Hence, the resistance of the prelithiated battery increases faster than that of the conventional one. Moreover, the irreversible loss of active Li ions during the charge and discharge process is effectively compensated by the additional lithium introduced at the early stage. Therefore, the capacity attenuation phenomena are obscured. The capacity change no longer reflects the side reaction status inside the battery.

Overall, customized special SOC estimation and SOH prediction algorithms are required. The algorithm for prelithiated LIBs with different prelithiation degrees is not universal.

## 4.6 | Engineering implementation challenges

The various prelithiation technologies mentioned above have their respective advantages. According to the mechanisms of different prelithiation methods, battery manufacturers may need a new production process and additional equipment, and a change in the environmental parameters of the factory to realize industrial prelithiation. Therefore, even though many prelithiation methods have been developed in the last decade, we still face a big challenge in industrial large-scale applications, and the challenge mainly comes from the following aspects:

1. Environmental control: For prelithiation technologies based on highly reactive lithium sources, such as SLMP, lithium metal foil, and some lithium organic compounds, the prelithiation process is very sensitive to moisture and oxygen. Therefore, it is necessary to eliminate moisture and oxygen in the atmosphere. Moreover, in most cases, the obtained lithiated electrodes are still sensitive to humidity and oxygen. Therefore, highly strict humidity and oxygen control are needed in all of the consequential working steps before the cell is sealed. For chemical and electrochemical prelithiation, additional production processes and equipment must be set up to eliminate the residual lithiation reagent and solvent on the lithiated electrode, which would not only increase the fabrication cost but also lengthen the production period of the LIBs.
2. Homogenization of the prelithiation degree: The uneven distribution of the lithiation degree on the electrode would induce many problems, including local overcharge, lithium plating, SEI defects, and so on. The inhomogeneity of the lithiation degree may be originated from material-level nonuniformity, electrode processing defects, fluctuations in the lithium foil thickness, prelithiation equipment faults, and some environmental

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

issues. It should be noted that substandard environmental control is an important issue to cause an inhomogeneous distribution of prelithiation; because if some part of the lithiated electrode is exposed to moisture, the added lithium in this part would lose its activity. To ensure the homogenization of the prelithiation degree, it is necessary to check all of the possible reasons and eliminate potential hazards.

3. Self-heating during electrolyte wetting: For the two most viable lithium prelithiation methods, the roll-to-roll method and SLMP method, the prelithiated lithium stays on the surface of the anode in metal or stable compound states till the electrolyte is filled. As shown in Figure 11C, lithium will start to react with the anode once it contacts the electrolyte and releases a mass of heat in a short time. This mechanism would form a high-temperature area on the surface of the anode, cause the failure of the active materials and electrolytes or even lead to thermal runaway of the battery. Hence, the electrolyte filling process of the prelithiated battery is different from that of the normal one. The extra device needs to be designed in the prelithiated battery production line to control the electrolyte filling speed precisely and dissipate the heat, especially for large cells.
4. SEI formation condition change: As mentioned above, the SEI formation process in prelithiated LIBs is different from that in conventional LIBs. For anode-side prelithiation, the SEI is formed during the prelithiation process instead of the first charge process. However, the as-formed SEI may be damaged during the rinsing, drying, rolling, and cell assembly processes. The purpose of the formation process may change from SEI formation to SEI repair. Correspondingly, the parameters of the formation process need to be adjusted according to the SEI quality and deterioration degree during previous steps. On the other hand, for cathode side prelithiation, the SEI is still formed during the first charge. However, the decomposition of cathode-side prelithiation reagents is usually accompanied by a large amount of gas release. Therefore, it is necessary to use specialized formation equipment with effective gas elimination accessories.

## 5 | CONCLUSIONS AND PERSPECTIVE

Prelithiation is one of the most feasible routes to upgrade the current LIB industry. This not only increases the reversible capacity but also raises the full-cell voltage, improves the deep discharge endurance, and elongates the cycle life. Meanwhile, the post-lithium-ion batteries (i.e., lithium-sulfur, lithium-oxygen, solid-state lithium metal, sodium-ion batteries) face the same problems like low ICE and specific energy. We believe that prelithiation treatment will become an indispensable step during these post-lithium-ion battery fabrication processes, and prelithiation technologies will offer reference significance and bring new insights for developing post-lithium-ion batteries. Furthermore, it is especially suitable for the next-generation large-capacity anode materials, such as Si/C and $SiO_x$/C, to compensate for the active lithium loss in the initial cycle.

At present, the industrialization of prelithiation has just begun in the pilot stage. The mechanical rolling method has shown advantages in lithiation efficiency and industrial feasibility and thus is mostly chosen by LIB manufacturers. However, this method still suffers from lithiation degree fluctuations in local areas. Many new prelithiation methods are emerging in laboratories, including prelithiation methods during the stages of material preparation, electrode processing, and cell assembly. These methods have shown potential in terms of lithiation controllability, cost, and equipment compatibility. However, further research is still needed to demonstrate industrial feasibility.

In addition to the advantages, prelithiation may also cause some problems. The anode materials in prelithiated cells always work at low potential, which leads to SEI formation condition change, high lithium plating possibility, and thermal runaway risk. In the manufacturing process, it is still difficult to precisely control the degree and uniformity of prelithiation, which brings unfavorable side reactions to local areas during operation. After the batteries are retired, it is necessary to consider the risk induced by the remaining high activity of lithiated anodes when disassembling the batteries.

In the future, with the promotion of next-generation large-capacity anode materials, prelithiation technology will become increasingly necessary. To further propel the technical progress of prelithiation, on one hand, it is worthwhile to explore brand new physical/chemical prelithiation methods, such as electrospray, plasma spray, or to develop new lithiation reagents; on the other hand, it is also important to improve the current prelithiation methods via optimization of processing details. For example, reducing the thickness of lithium metal foil is of great importance for mechanical rolling prelithiation.

In addition to prelithiation methods, we should also note that prelithiated LIBs have significantly different electrochemical behaviors from conventional LIBs. It is necessary to develop corresponding cell design principles, battery management strategies, and battery recycling techniques. At present, there are very few publications

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

on these topics. However, we believe that more and more advanced technologies can take high-performance pre-lithiated LIBs with maximum advantages to practical applications.

## ACKNOWLEDGMENTS

The authors acknowledge the support of the National Natural Science Foundation of China (Nos. 22179045 and 5202780089).

## CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

## ORCID

*Yunhui Huang* https://orcid.org/0000-0001-8286-3915

## REFERENCES

1. Nathaniel SP, Alam MS, Murshed M, Mahmood H, Ahmad P. The roles of nuclear energy, renewable energy, and economic growth in the abatement of carbon dioxide emissions in the G7 countries. *Environ Sci Pollut Res*. 2021;28(35):47957-47972.
2. Bhattacharya M, Paramati SR, Ozturk I, Bhattacharya S. The effect of renewable energy consumption on economic growth: evidence from top 38 countries. *Appl Energy*. 2016;162:733-741.
3. Goodenough JB, Park KS. The i-ion rechargeable battery: a perspective. *J Am Chem Soc*. 2013;135(4):1167-1176.
4. Yoshino A. The birth of the lithium-ion battery. *Angew Chem Int Ed*. 2012;51(24):5798-5800.
5. Kittner N, Lill F, Kammen DM. Energy storage deployment and innovation for the clean energy transition. *Nat Energy*. 2017;2(9):17125.
6. Armand M, Tarascon JM. Building better batteries. *Nature*. 2008;451(7179):652-657.
7. Lu J, Chen Z, Pan F, Cui Y, Amine K. High-performance anode materials for rechargeable lithium-ion batteries. *Electrochem Energy Rev*. 2018;1(1):35-53.
8. Scrosati B, Garche J. Lithium batteries: status, prospects and future. *J Power Sources*. 2010;195(9):2419-2430.
9. Yan Z, Jin H, Guo J. Low-temperature synthesis of graphitic carbon-coated silicon anode materials. *Carbon Energy*. 2019;1(2):246-252.
10. Wu J, Cao Y, Zhao H, Mao J, Guo Z. The critical role of carbon in marrying silicon and graphite anodes for high-energy lithium-ion batteries. *Carbon Energy*. 2019;1(1):57-76.
11. Wang F, Chen G, Zhang N, Liu X, Ma R. Engineering of carbon and other protective coating layers for stabilizing silicon anode materials. *Carbon Energy*. 2019;1(2):219-245.
12. Mahlia T, Saktisahdan T, Jannifar A, Hasan M, Matseelar H. A review of available methods and development on energy storage; technology update. *Renewable Sustainable Energy Rev*. 2014;33:532-545.
13. Suberu MY, Mustafa MW, Bashir N. Energy storage systems for renewable energy power sector integration and mitigation of intermittency. *Renewable Sustainable Energy Rev*. 2014;35:499-514.
14. Goodenough JB. Cathode materials: a personal perspective. *J Power Sources*. 2007;174(2):996-1000.
15. Manthiram A, Goodenough JB. Layered lithium cobalt oxide cathodes. *Nat Energy*. 2021;6(3):323.
16. Dou F, Shi L, Chen G, Zhang D. Silicon/carbon composite anode materials for lithium-ion batteries. *Electrochem Energy Rev*. 2019;2(1):149-198.
17. Whitehead AH, Schreiber M. Current collectors for positive electrodes of lithium-based batteries. *J Electrochem Soc*. 2005;152(11):A2105-A2113.
18. Zheng F, Kotobuki M, Song S, Lai MO, Lu L. Review on solid electrolytes for all-solid-state lithium-ion batteries. *J Power Sources*. 2018;389:198-213.
19. An SJ, Li J, Daniel C, Mohanty D, Nagpure S, Wood DL. The state of understanding of the lithium-ion battery graphite solid electrolyte interphase (SEI) and its relationship to formation cycling. *Carbon*. 2016;105:52-76.
20. Lu P, Li C, Schneider EW, Harris SJ. Chemistry, impedance, and morphology evolution in solid electrolyte interphase films during formation in lithium ion batteries. *J Phys Chem C*. 2014;118(2):896-903.
21. Yang Y, Chong Y, Huang J. Research progress of solid electrolyte interphase in lithium batteries. *Acta Phys-Chim Sin*. 2021;37(11):62-74.
22. Holtstiege F, Wilken A, Winter M, Placke T. Running out of lithium? A route to differentiate between capacity losses and active lithium losses in lithium-ion batteries. *Phys Chem Chem Phys*. 2017;19(38):25905-25918.
23. Forney MW, Ganter MJ, Staub JW, Ridgley RD, Landi BJ. Prelithiation of silicon-carbon nanotube anodes for lithium ion batteries by stabilized lithium metal powder (SLMP). *Nano Lett*. 2013;13(9):4158-4163.
24. Sun C, Zhang X, Li C, Wang K, Sun X, Ma Y. Recent advances in prelithiation materials and approaches for lithium-ion batteries and capacitors. *Energy Storage Mater*. 2020;32:497-516.
25. Wang F, Wang B, Li J, et al. Prelithiation: a crucial strategy for boosting the practical application of next-generation lithium ion battery. *ACS Nano*. 2021;15(2):2197-2218.
26. Zhan R, Wang X, Chen Z, Seh ZW, Wang L, Sun Y. Promises and challenges of the practical implementation of prelithiation in lithium-ion batteries. *Adv Energy Mater*. 2021;11(35):2101565.
27. Pan Q, Zuo P, Mu T, et al. Improved electrochemical performance of micro-sized SiO-based composite anode by prelithiation of stabilized lithium metal powder. *J Power Sources*. 2017;347:170-177.
28. Domi Y, Usui H, Iwanari D, Sakaguchi H. Effect of mechanical pre-lithiation on electrochemical performance of silicon negative electrode for lithium-ion batteries. *J Electrochem Soc*. 2017;164(7):A1651-A1654.
29. Costard J, Ender M, Weiss M, Ivers E. Three-electrode setups for lithium-ion batteries. *J Electrochem Soc*. 2016;164(2):A71-A80.
30. Liao X, Ding Z, Yin Z. Excellent performance of a modified graphite anode for lithium-ion battery application. *Ionics*. 2020;26(11):5367-5373.
31. Kasnatscheew J, Rodehorst U, Streipert B, et al. Learning from overpotentials in lithium ion batteries: a case study on the $LiNi_{1/3}Co_{1/3}Mn_{1/3}O_2$ (NCM) cathode. *J Electrochem Soc*. 2016;163(14):A2943-A2950.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

32. Yoshio M, Wang H, Fukuda K, Umeno T, Dimov N, Ogumi Z. Carbon-coated Si as a lithium-ion battery anode material. *J Electrochem Soc*. 2002;149(12):A1598-A1603.
33. Chae S, Ko M, Kim K, Ahn K, Cho J. Confronting issues of the practical implementation of i anode in high-energy lithium-ion batteries. *Joule*. 2017;1(1):47-60.
34. Kierzek K, Machnikowski J. Factors influencing cycle-life of full i-ion cell built from Si/C composite as anode and conventional cathodic material. *Electrochim Acta*. 2016;192:475-481.
35. Klett M, Gilbert JA, Trask SE, et al. Electrode behavior RE-visited: monitoring potential windows, capacity loss, and impedance changes in $Li_{1.03}(Ni_{0.5}Co_{0.2}Mn_{0.3})_{0.97}O_2$/silicon-graphite full cells. *J Electrochem Soc*. 2016;163(6):A875-A887.
36. Mao Y, Wang X, Xia S, et al. High-voltage charging-induced strain, heterogeneity, and micro-cracks in secondary particles of a nickel-rich layered cathode material. *Adv Funct Mater*. 2019;29(18):1900247.
37. Spotnitz R. Simulation of capacity fade in lithium-ion batteries. *J Power Sources*. 2003;113(1):72-80.
38. Tipton WW, Bealing CR, Mathew K, Hennig RG. Structures, phase stabilities, and electrical potentials of Li-Si battery anode materials. *Phys Rev B*. 2013;87(18):184114.
39. Li Z, Zhao H, Lv P, et al. Watermelon-like structured $SiO_x$-$TiO_2$@C nanocomposite as a high-performance lithium-ion battery anode. *Adv Funct Mater*. 2018;28(31):1605711.
40. Reddy MV, Subba RG, Chowdari B. Metal oxides and oxysalts as anode materials for Li ion batteries. *Chem Rev*. 2013;113(7):5364-5457.
41. Wang A, Kadam S, Li H, Shi S, Qi Y. Review on modeling of the anode solid electrolyte interphase (SEI) for lithium-ion batteries. *npj Comput Mater*. 2018;4(1):1-26.
42. Michalak B, Berkes BB, Sommer H, Brezesinski T, Janek J. Electrochemical Cross-talk leading to gas evolution and capacity fade in $LiNi_{0.5}Mn_{1.5}O_4$/Graphite Full-cells. *J Phys Chem C*. 2016;121(1):211-216.
43. Zhang X, inventor; Contemporary Amperex Technology Co Ltd, assignee. Split rolling one-time lamination lithium supplement device. CN patent 105,845,896. May 16, 2016.
44. Zhang X, inventor; Contemporary Amperex Technology Co Ltd, assignee. Split-rolling and split-combining lithium supplement device. CN patent 105,845,897. May 16, 2016.
45. Chen N, Gu H, Shi D, Jin H, Li Z, Zhao Q, inventors; Contemporary Amperex Technology Co Ltd, assignee. Lithium supplementing agent, positive pole piece, isolating membrane and lithium ion battery. CN patent 111,384,428. December 29, 2018.
46. Lee SK, Choi BK, Lee BB, Jeong BH, Kim KH, Kim JB, inventors; LG Chemical Ltd, assignee. Method for prelithiation, method for fabricating lithium secondary battery comprising the method, and lithium secondary battery fabricated from the fabricating method. CN patent 104,584,278. April 11, 2014.
47. Zhang G, Bi G, Feng D, inventors; Huawei Digital Tech Suzhou Co Ltd, assignee. Lithium supplement method and device of battery. CN patent 107,978,790. Novement 16, 2017.
48. Zhao W, Li S, Yuan H, Tang W, Li J, Xu Y, inventors; Zhuhai Cosmx Battery Co Ltd, assignee. Lithium supplement device and lithium supplement method for negative electrode sheets of lithium-ion batteries. CN patent 110,459,731. August 17, 2019.
49. Gao H, Zhang Y, Yuan Z, Liu J, inventors; Huizhou Eve Energy Co Ltd, assignee. Preparation method of lithium-ion battery based on lithium intercalation negative electrode. CN patent 109,346,665. August 27, 2018.
50. Wen B, Sun Q, Li X, Wang Y, Liu J, inventors; Citic Guoan Mgl Power Science & Tech Co Ltd, assignee. Lithium refilling electrolyte and lithium refilling method for lithium ion battery. CN patent 108,878,974. May 16, 2017.
51. Xu TAO, Wang Cinventors; Hefei Guoxuan High-Tech Power Energy Co Ltd, assignee. Method for supplementing lithium to negative pole piece. CN patent 108,899,468. June 27, 2018.
52. Yang M, inventor; Hefei Guoxuan High-Tech Power Energy Co Ltd, assignee. Pre-lithiation method and device for negative pole pieces of lithium-ion battery. CN patent 105,845,894. May 4, 2016.
53. Peramunage D, Abraham K. Preparation and electrochemical characterization of overlithiated spinel $LiMn_2O_4$. *J Electrochem Soc*. 1998;145(4):1131-1136.
54. Liu X, Tan Y, Wang W, et al. Conformal prelithiation nanoshell on $LiCoO_2$ enabling high-energy lithium-ion batteries. *Nano Lett*. 2020;20(6):4558-4565.
55. Sun Y, Li Y, Sun J, Li Y, Pei A, Cui Y. Stabilized $Li_3N$ for efficient battery cathode prelithiation. *Energy Storage Mater*. 2017;6:119-124.
56. Sun Y, Lee HW, Seh ZW, et al. High-capacity battery cathode prelithiation to offset initial lithium loss. *Nat Energy*. 2016;1(1):15008.
57. Shanmukaraj D, Grugeon S, Laruelle S, Douglade G, Tarascon JM, Armand M. Sacrificial salts: compensating the initial charge irreversibility in lithium batteries. *Electrochem Commun*. 2010;12(10):1344-1347.
58. Park H, Yoon T, Kim YU, Ryu JH, Oh SM. $Li_2NiO_2$ as a sacrificing positive additive for lithium-ion batteries. *Electrochim Acta*. 2013;108:591-595.
59. Su X, Lin C, Wang X, et al. A new strategy to mitigate the initial capacity loss of lithium ion batteries. *J Power Sources*. 2016;324:150-157.
60. Zhang L, Dose WM, Vu AD, Johnson CS, Lu W. Mitigating the initial capacity loss and improving the cycling stability of silicon monoxide using $Li_5FeO_4$. *J Power Sources*. 2018;400:549-555.
61. Wang D, Zhang Z, Hong B, Lai Y. Self-sacrificial organic lithium salt enhanced initial Coulombic efficiency for safer and greener lithium-ion batteries. *Chem Commun*. 2019;55(72):10737-10739.
62. Shen Y, Qian J, Yang H, Zhong F, Ai X. Chemically prelithiated Hard-carbon anode for high power and high capacity Li-ion batteries. *Small*. 2020;16(7):1907602.
63. Yan MY, Li G, Zhang J, et al. Enabling $SiO_x$/C anode with high initial coulombic efficiency through a chemical Pre-lithiation strategy for high-energy-density lithium-ion batteries. *ACS Appl Mater Interfaces*. 2020;12(24):27202-27209.
64. Zhang X, Qu H, Ji W, et al. An electrode-level prelithiation of SiO anodes with organolithium compounds for lithium-ion batteries. *J Power Sources*. 2020;478:229067.
65. Yoshida S, Masuo Y, Shibata D, Haruta M, Doi T, Inaba M. Li pre-doping of amorphous silicon electrode in li-naphthalene complex solutions. *Electrochemistry*. 2015;83(10):843-845.


26373968, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

66. Jarvis C, Lain M, Yakovleva M, Gao Y. A prelithiated carbon anode for lithium-ion battery applications. *J Power Sources*. 2006;162(2):800-802.
67. Zhou D, Liu M, Yun Q, et al. A novel lithiated Silicon–Sulfur battery exploiting an optimized Solid-Like electrolyte to enhance safety and cycle life. *Small*. 2017;13(3):1602015.
68. Marinaro M, Weinberger M, Wohlfahrt-Mehrens M. Toward pre-lithiatied high areal capacity silicon anodes for lithium-ion batteries. *Electrochim Acta*. 2016;206:99-107.
69. Wang Z, Fu Y, Zhang Z, et al. Application of stabilized lithium metal powder (SLMP®) in graphite anode-a high efficient prelithiation method for lithium-ion batteries. *J Power Sources*. 2014;260:57-61.
70. Zhao J, Lu Z, Liu N, Lee HW, Mcdowell MT, Cui Y. Dry-air-stable lithium silicide-lithium oxide core-shell nanoparticles as high-capacity prelithiation reagents. *Nat Commun*. 2014;5:5088.
71. Zhao J, Lu Z, Wang H, et al. Artificial solid electrolyte interphase-protected $Li_xSi$ nanoparticles: an efficient and stable prelithiation reagent for lithium-ion batteries. *J Am Chem Soc*. 2015;137(26):8372-8375.
72. Li S, Wang C, Yu J, Han Y, Lu Z. Understanding the role of conductive polymer in thermal lithiation and battery performance of Li-Sn alloy anode. *Energy Storage Mater*. 2019;20:7-13.
73. Alaboina PK, Cho JS, Uddin MJ, Cho SJ. Mechanically prelithiated silicon nano alloy as highly engineered anode material. *Electrochim Acta*. 2017;258:623-630.
74. Li Z, Zhang Y, Liu T, et al. Silicon anode with high initial Coulombic efficiency by modulated trifunctional binder for high-areal-capacity lithium-ion batteries. *Adv Energy Mater*. 2020;10(20):1903110.
75. Vadivel S, Sawangphruk M. Prelithiated perfluoro-ionomer as an alternative binder for the state-of-the-art ni-rich $LiNi_{0.8}Co_{0.15}Al_{0.05}O_2$ cathode of next-generation lithium-ion batteries. *J Mater Chem A*. 2020;8(39):20714-20724.
76. Rezqita A, Kathribail AR, Kahr J, Jahn M. Analysis of degradation of Si/Carbon|| $LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$ full cells: effect of prelithiation. *J Electrochem Soc*. 2019;166(3):A5483-A5488.
77. Kim HJ, Choi S, Lee SJ, et al. Controlled prelithiation of silicon monoxide for high performance lithium-ion rechargeable full cells. *Nano Lett*. 2016;16(1):282-288.
78. Meng Q, Li G, Yue J, Xu Q, Yin YX, Guo YG. High-Performance lithiated $SiO_x$ anode obtained by a controllable and efficient prelithiation strategy. *ACS Appl Mater Interfaces*. 2019;11(35):32062-32068.
79. Lee DI, Yang HW, Kang WS, Kim J, Kim SJ. Optimal condition of solid-electrolyte-interphase prepared by controlled prelithiation for high performance Li-ion batteries. *J Electrochem Soc*. 2019;166(4):A787-A792.
80. Liu N, Hu L, Mcdowell MT, Jackson A, Cui Y. Prelithiated silicon nanowires as an anode for lithium ion batteries. *ACS Nano*. 2011;5(8):6487-6493.
81. Zhou H, Wang X, Chen D. Li-Metal-Free prelithiation of Si-based negative electrodes for full Li-ion batteries. *ChemSusChem*. 2015;8(16):2737-2744.
82. Hassoun J, Lee KS, Sun YK, Scrosati B. An advanced lithium ion battery based on high performance electrode materials. *J Am Chem Soc*. 2011;133(9):3139-3143.
83. Winter M, Barnett B, Xu K. Before Li ion batteries. *Chem Rev*. 2018;118(23):11433-11456.
84. De La Llave E, Borgel V, Park KJ, et al. Comparison between Na-ion and Li-ion cells: understanding the critical role of the cathodes stability and the anodes pretreatment on the cells behavior. *ACS Appl Mater Interfaces*. 2016;8(3):1867-1875.
85. Holtstiege F, Koç T, Hundehege T, Siozios V, Winter M, Placke T. Toward high power batteries: pre-lithiated carbon nanospheres as high rate anode material for lithium ion batteries. *ACS Appl Energy Mater*. 2018;1(8):4321-4331.
86. Zhang J, Wu H, Wang J, Shi J, Shi Z. Pre-lithiation design and lithium ion intercalation plateaus utilization of meso-carbon microbeads anode for lithium-ion capacitors. *Electrochim Acta*. 2015;182:156-164.
87. Guo A, Chen E, Heller A, Mullins CB. Controlled prelithiation of PbS to $Pb/Li_2S$ for high initial Coulombic efficiency in lithium ion batteries. *J Electrochem Soc*. 2019;166(10):A1939-A1943.
88. Sivakkumar SR, Pandolfo A. Evaluation of lithium-ion capacitors assembled with pre-lithiated graphite anode and activated carbon cathode. *Electrochim Acta*. 2012;65:280-287.
89. Decaux C, Lota G, Raymundo E, Frackowiak E, Béguin F. Electrochemical performance of a hybrid lithium-ion capacitor with a graphite anode preloaded from lithium bis(trifluoromethane) sulfonimide-based electrolyte. *Electrochim Acta*. 2012;86:282-286.
90. Wu Z, Ji S, Hu Z, et al. Pre-Lithiation of $Li(Ni_{1-x-y}Mn_xCo_y)O_2$ materials enabling enhancement of performance for Li-Ion battery. *ACS Appl Mater Interfaces*. 2016;8(24):15361-15368.
91. Aravindan V, Nan S, Keppeler M, Madhavi S. Pre-lithiated $Li_xMn_2O_4$: A new approach to mitigate the irreversible capacity loss in negative electrodes for Li-ion battery. *Electrochim Acta*. 2016;208:225-230.
92. Wang G, Li F, Liu D, et al. Chemical prelithiation of negative electrodes in ambient air for advanced lithium-ion batteries. *ACS Appl Mater Interfaces*. 2019;11(9):8699-8703.
93. Zhang X, Qu H, Ji W, et al. Fast and controllable prelithiation of hard carbon anodes for lithium-ion batteries. *ACS Appl Mater Interfaces*. 2020;12(10):11589-11599.
94. Shen Y, Zhang J, Pu Y, et al. Effective chemical prelithiation strategy for building a silicon/sulfur Li-ion battery. *ACS Energy Lett*. 2019;4(7):1717-1724.
95. Li F, Wang G, Zheng D, et al. Controlled prelithiation of $SnO_2$/C nanocomposite anodes for building full lithium-ion batteries. *ACS Appl Mater Interfaces*. 2020;12(17):19423-19430.
96. Wang G, Li F, Liu D, et al. High performance lithium-ion and lithium–sulfur batteries using prelithiated phosphorus/carbon composite anode. *Energy Storage Mater*. 2020;24:147-152.
97. Liu Z, Ma S, Mu X, Li R, Yin G, Zuo P. A scalable cathode chemical prelithiation strategy for advanced silicon-based lithium ion full batteries. *ACS Appl Mater Interfaces*. 2021;13(10):11985-11994.
98. Shen Y, Shen X, Yang M, et al. Achieving desirable initial Coulombic efficiencies and full capacity utilization of Li-ion batteries by chemical prelithiation of graphite anode. *Adv Funct Mater*. 2021;31(24):2101181.
99. Choi J, Jeong H, Jang J, et al. Weakly solvating solution enables chemical prelithiation of graphite-$SiO_x$ anodes for high-energy Li-ion batteries. *J Am Chem Soc*. 2021;143(24):9169-9176.

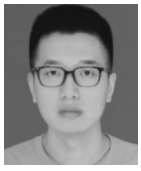
26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

100. Jang J, Kang I, Choi J, et al. Molecularly tailored lithium-arene complex enables chemical prelithiation of high-capacity lithium-ion battery anodes. *Angew Chem Int Ed*. 2020;132(34):14581-14588.
101. Chen H, Yang Y, Boyle DT, et al. Free-standing ultrathin lithium metal-graphene oxide host foils with controllable thickness for lithium batteries. *Nat Energy*. 2021;6(8):790-798.
102. Xu H, Li S, Zhang C, et al. Roll-to-roll prelithiation of sn foil anode suppresses gassing and enables stable full-cell cycling of lithium ion batteries. *Energy Environ Sci*. 2019;12(10):2991-3000.
103. Xu H, Li S, Chen X, et al. Sn-Alloy foil electrode with mechanical prelithiation: full-cell performance up to 200 cycles. *Adv Energy Mater*. 2019;9(42):1902150.
104. Fan H, Chen B, Li S, et al. Nanocrystalline Li-Al-Mn-Si foil as reversible li host: electronic percolation and electrochemical cycling stability. *Nano Lett*. 2019;20(2):896-904.
105. Hu Z, Chen S, Gong Z, Li W, Xie B, inventors; Contemporary Amperex Technology Co Ltd, assignee. Pole piece lithium supplementing device. CN patent 209,401,737. January 30, 2019.
106. Li K, Wu Z, Zhang W, inventors; Contemporary Amperex Technology Co Ltd, assignee. Lithium supplement method and system of pole plate. CN patent 105,489,846. January 11, 2016.
107. Yue XY, Yao YX, Zhang J, et al. Unblocked electron channels enable efficient contact prelithiation for Lithium-Ion batteries. *Adv Mater*. 2022;34(15):2110337.
108. Kim K, Shon J, Jeong H, Park H, Lim SJ, Heo J. Improving the cyclability of silicon anodes for lithium-ion batteries using a simple pre-lithiation method. *J Power Sources*. 2020;459:228066.
109. Takezawa H, Ito S, Yoshizawa H, Abe T. Electrochemical properties of a $SiO_x$ film anode pre-lithiated by evaporation of metallic Li in Li-ion batteries. *Chem Lett*. 2017;46(9):1365-1367.
110. Yang SY, Yue XY, Xia HY, et al. Battery prelithiation enabled by lithium fixation on cathode. *J Power Sources*. 2020;480:229109.
111. Chen Q, Wu X, Hu P, Fang B, Liu J, inventors; Lanxi Zhide New Energy Material Co Ltd, assignee. Device and method for filling lithium for anode and cathode materials of lithium ion battery through chemical vapor deposition (CVD). CN patent 110,438,469. August 8, 2019.
112. Watanabe T, Tsuda T, Ando N, et al. An improved pre-lithiation of graphite anodes using through-holed cathode and anode electrodes in a laminated lithium ion battery. *Electrochim Acta*. 2019;324:134848.
113. Zensen N. Development of a prelithiation method on cell stack level for implementing silicon anodes. Doctoral dissertation. Universität Ulm; 2020.
114. Sun C, Zhang X, Li C, Wang K, Sun X, Ma Y. High-efficiency sacrificial prelithiation of lithium-ion capacitors with superior energy-storage performance. *Energy Storage Mater*. 2020;24:160-166.
115. Diaz-Lopez M, Chater PA, Bordet P, et al. $Li_2O$: Li-Mn-O disordered Rock-salt nanocomposites as cathode prelithiation additives for High-energy density Li-ion batteries. *Adv Energy Mater*. 2020;10(7):1902788.
116. Jeżowski P, Fic K, Crosnier O, Brousse T, Béguin F. Use of sacrificial lithium nickel oxide for loading graphitic anode in Li-ion capacitors. *Electrochim Acta*. 2016;206:440-445.
117. Park MS, Lim YG, Hwang SM, et al. Scalable integration of $Li_5FeO_4$ towards robust, high-performance lithium-ion hybrid capacitors. *ChemSusChem*. 2014;7(11):3138-3144.
118. Vitins G, Raekelboom E, Weller M, Owen J. $Li_2CuO_2$ as an additive for capacity enhancement of lithium ion cells. *J Power Sources*. 2003;119:938-942.
119. Noh M, Cho J. Role of $Li_6CoO_4$ cathode additive in i-ion cells containing low Coulombic efficiency anode material. *J Electrochem Soc*. 2012;159(8):A1329-A1334.
120. Lin X, Khosravinia K, Hu X, Li J, Lu W. Lithium plating mechanism, detection, and mitigation in lithium-Ion batteries. *Prog Energy Combust Sci*. 2021;87:100953.
121. Chandrasekaran R. Quantification of bottlenecks to fast charging of lithium-ion-insertion cells for electric vehicles. *J Power Sources*. 2014;271:622-632.
122. Hein S, Latz A. Influence of local lithium metal deposition in 3D microstructures on local and global behavior of lithium-ion batteries. *Electrochim Acta*. 2016;201:354-365.
123. Wang H, Zhu Y, Kim SC, et al. Underpotential lithium plating on graphite anodes caused by temperature heterogeneity. *Proc Natl Acad Sci U S A*. 2020;117(47):29453-29461.
124. Harris SJ, Timmons A, Baker DR, Monroe C. Direct in situ measurements of Li transport in Li-ion battery negative electrodes. *Chem Phys Lett*. 2010;485(4-6):265-274.
125. Ahmed S, Bloom I, Jansen AN, et al. Enabling fast charging—a battery technology gap assessment. *J Power Sources*. 2017;367:250-262.
126. Legrand N, Knosp B, Desprez P, Lapicque F, Raël S. Physical characterization of the charging process of a Li-ion battery and prediction of Li plating by electrochemical modelling. *J Power Sources*. 2014;245:208-216.
127. Li Z, Huang J, Liaw BY, Metzler V, Zhang J. A review of lithium deposition in lithium-ion and lithium metal secondary batteries. *J Power Sources*. 2014;254:168-182.
128. Ecker M, Sabet PS, Sauer DU. Influence of operational condition on lithium plating for commercial lithium-ion batteries—electrochemical experiments and post-mortem-analysis. *Appl Energy*. 2017;206:934-946.
129. Wu X, Song B, Chien PH, et al. Structural evolution and transition dynamics in lithium ion battery under fast charging: an operando neutron diffraction investigation. *Adv Sci*. 2021;8(21):2102318.
130. Feng X, Zheng S, Ren D, et al. Investigating the thermal runaway mechanisms of lithium-ion batteries based on thermal analysis database. *Appl Energy*. 2019;246:53-64.
131. Wang Q, Sun J, Yao X, Chen C. Thermal behavior of lithiated graphite with electrolyte in lithium-ion batteries. *J Electrochem Soc*. 2005;153(2):A329-A333.
132. Liu X, Yin L, Ren D, et al. In situ observation of thermal-driven degradation and safety concerns of lithiated graphite anode. *Nat Commun*. 2021;12:4235.
133. Rao Z, Wang S, Zhang G. Simulation and experiment of thermal energy management with phase change material for ageing $LiFePO_4$ power battery. *Energy Convers Manag*. 2011;52(12):3408-3414.
134. Chiasson J, Vairamohan B. Estimating the state of charge of a battery. *IEEE Trans Control Syst Technol*. 2003;4:2863-2868.
135. Chen N, Zhang P, Dai J, Gui W. Estimating the state-of-charge of lithium-ion battery using an H-infinity observer

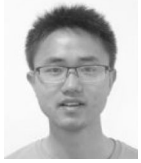

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

based on electrochemical impedance model. *IEEE Access*. 2020;8:26872-26884.
136. Huet F. A review of impedance measurements for determination of the state-of-charge or state-of-health of secondary batteries. *J Power Sources*. 1998;70(1):59-69.
137. Ai G, Wang Z, Zhao H, et al. Scalable process for application of stabilized lithium metal powder in Li-ion batteries. *J Power Sources*. 2016;309:33-41.
138. Hu X, Xu L, Lin X, Pecht M. Battery lifetime prognostics. *Joule*. 2020;4(2):310-346.
139. Aragon Zulke A, Hoster H, Mercer M, et al. Cells on idle: anode versus cathode driven side reactions. *Batter Supercaps*. 2021;4(6):934-947.

## AUTHOR BIOGRAPHIES



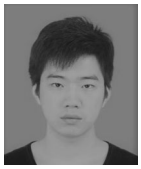
**Zhenyu Huang** received his M.S. degree in 2016 from Nanchang University. He joined the Huazhong University of Science and Technology-Wuxi research center in 2017 and Jiangsu Jitri-Hust Intelligent Equipment Technology Co., Ltd. in 2019. Huang's research interests are mainly focused on the application of ultrasonic technology in lithium-ion batteries and ultrasonic data analysis.



**Zhe Deng** received his Ph.D. degree in 2021 from Huazhong University of Science and Technology. He worked in the Huazhong University of Science and Technology-Wuxi research center. His research interests are mainly focused on the advanced in situ characterization of lithium batteries and the application of ultrasonic technology in the research of lithium-ion batteries.



**Yun Zhong** received her Ph.D. degree in 2022 from Huazhong University of Science and Technology. Her current research focuses on aqueous zinc-ion batteries.



**Mingkang Xu** obtained a master's degree in engineering from Huazhong University of science and technology. During his postgraduate study, he has been committed to the research and exploration of ultrasonic scanning technology of lithium-ion battery, and after graduation, he worked at Huazhong University of science and technology-Wuxi Research Institute to further carry out customized research on the technology in combination with industrial pain points.



**Sida Li** received his master's degree in 2022 from Huazhong University of Science and Technology. He worked in State Key Laboratory of Material Processing and Die & Mould Technology, School of Materials Science and Engineering. His research interests are mainly focused on construction of stable interface in batteries and development of new self-stratified batteries.



**Xueting Liu** received her MS degree in 2021 from Huazhong University of Science and Technology. She worked in the Huazhong University of Science and Technology-Wuxi research center. Her current research focuses on the application of ultrasonic technology in the research of lithium-ion batteries.



**Yu Zhou** received his M.S. degree in Material engineering from Huazhong University of Science and Technology in 2019. He joined the Huazhong University of Science and Technology-Wuxi research center in March 2018. His research interests are mainly focused on ultrasonic data analysis and the application of ultrasonic technology in lithium-ion batteries.



**Kai Huang** received his M.S. degree in Material engineering from Huazhong University of Science and Technology in 2019. He joined the Huazhong University of Science and Technology-Wuxi Research Institute in June 2018. His research interest mainly focuses on the interface evolution of lithium-ion battery by ultrasonic technology.



**Yue Shen** received his Ph.D. degree in 2011 from Peking University. He was a visiting student at the Georgia Institute of Technology from 2008 to 2010. He is now an associate professor at Huazhong University of Science and Technology. His research interest mainly focuses on lithium rechargeable batteries.

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License



**Yunhui Huang** received his Ph.D. degree from Peking University. In 2000, he worked as a postdoctoral researcher at Peking University. From 2002 to 2004, he worked as an associate professor at the Fudan University and a JSPS fellow at the Tokyo Institute of Technology, Japan. He then worked with Prof. John B. Goodenough at the University of Texas at Austin for more than 3 years. In 2008, he became a chair professor of materials science at Huazhong University of Science and Technology. His research group works on batteries for energy storage and conversion.

**How to cite this article:** Huang Z, Deng Z, Zhong Y, et al. Progress and challenges of prelithiation technology for lithium-ion battery. *Carbon Energy*. 2022;4:1107-1132. doi:10.1002/cey2.256

26379368, 2022, 6, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/cey2.256 by Priyankar Biswas - TESTREGPROV3 , Wiley Online Library on [06/05/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License